A Framework for Incentivizing Client Participation and Connectivity in Decentralized Federated Learning

Susana Marques
INESC TEC
Braga, Portugal
susana.v.marques@inesctec.pt

Jivesh Jain
Carnegie Mellon University
Pittsburgh, U.S.A
jpjain@andrew.cmu.edu

Anupam Nayak
Carnegie Mellon University
Pittsburgh, U.S.A
anupamn@andrew.cmu.edu

Osman Yağan
Carnegie Mellon University
Pittsburgh, U.S.A
oyagan@andrew.cmu.edu

Fábio Coelho
INESC TEC
Braga, Portugal
fabio.a.coelho@inesctec.pt

Gauri Joshi
Carnegie Mellon University
Pittsburgh, U.S.A
gaurij@andrew.cmu.edu

Abstract

In decentralized federated learning, edge clients collaboratively train a model over a network via peer-to-peer communication without relying on a central server. Each client may have its own requirements, such as achieving a minimum training loss, which it expects the learned model to satisfy. Due to data heterogeneity, these requirements might not be met, leading some clients to opt out of collaborative training. More critically, if certain highly connected clients drop out, it can lead to network fragmentation, thus disrupting the training process and causing the remaining clients to lose access to valuable information from other parts of the network. In this work, we introduce a novel framework designed to incentivize the continued participation of key clients, where the notion of "importance" can be defined using various criteria (e.g., data quality metrics, connectivity metrics, etc). We establish theoretical convergence guarantees for our proposed algorithms and validate our approach through empirical studies across various network topologies, demonstrating that our solution outperforms standard decentralized learning under voluntary dropouts, achieving higher test-time accuracy while significantly improving client retention. Overall, our work yields accuracy improvements of up to 33% across settings, with client retention rates reaching above 99% in the best cases.

CCS Concepts

• **Computing methodologies → Machine learning**.

Keywords

Decentralized Learning, Federated Learning, Client incentive mechanisms

ACM Reference Format:
Susana Marques, Jivesh Jain, Anupam Nayak, Osman Yağan, Fábio Coelho, and Gauri Joshi. 2026. A Framework for Incentivizing Client Participation and Connectivity in Decentralized Federated Learning. In *Proceedings of (ACM MobiHoc '26).* ACM, New York, NY, USA, 35 pages. https://doi.org/XXXXXXX.XXXXXXX

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ACM MobiHoc '26, Tokyo, Japan
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ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Federated learning is a modern distributed machine learning paradigm [13, 22, 29] where edge clients such as cellphones or sensors collaboratively train a machine learning model using local datasets that cannot be transferred to the cloud for centralized training. In the standard implementation of federated learning, called cross-device federated learning, a central server aggregates updates from edge devices. In systems where there is no central trustworthy aggregator, also called cross-silo federated learning, the decentralized machine learning paradigm [16, 17, 23, 30, 31], nodes send and receive updates to and from neighboring nodes in a given network topology connecting them. Examples of such decentralized learning include banks, hospitals, or other entities that have local siloes of private data, but want to collaboratively learn models with other trusted entities. The network connecting the nodes then represents a trust graph used for communication with other nodes. Most previous works consider that all the clients participate in the network throughout the training process. In this paper, we study the problem of avoiding client dropouts, where nodes either voluntarily or involuntarily leave the collaborative training process.

Why would Clients Dropout from Decentralized Learning? In fully decentralized learning framework in a connected network topology, the training datasets $\mathcal{D}_i$ across clients can be highly heterogeneous, a situation frequently encountered in practical applications [13]. Data heterogeneity results in the empirical risk objective functions $F_i(\mathbf{x})$ induced by the datasets being different across clients, and also different from the global objective function $\sum_{i=1}^{N} F_i(\mathbf{x})/N$. Each client maintains its own version of the model $\mathbf{x}_i$ that is trained for a few stochastic gradient descent (SGD) iterations on its local dataset and then averaged with neighboring clients. After sufficiently many iterations of local training and aggregation with neighboring nodes in a connected topology, and under standard assumptions on the objective functions, all the nodes typically converge to a single collaboratively trained model $\mathbf{x}$. When the local data distribution is similar to the global distribution of the union of all the clients' datasets, the shared collaboratively trained model $\mathbf{x}$ generalizes better than a local model $\mathbf{x}_i$ trained by a client i in isolation using its limited local data $\mathcal{D}_i$. However, due to data heterogeneity, if the global distribution is significantly misaligned with a client's local

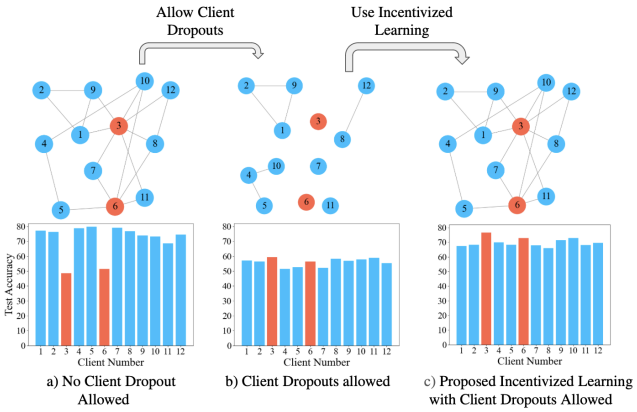


Figure 1: For a network of 12 clients, we compare the clients' test accuracies across three different algorithms: (a) decentralized learning without dropout results in low accuracy for clients 3 and 6; (b) allowing dissatisfied clients to dropout leads to disconnection and accuracy loss; (c) our proposed framework incentivizes clients to stay, preserving or improving accuracy for most clients.

data distribution, the client may prefer to leave the collaboration and train its own model in isolation.

Accuracy and Connectivity Disruption due to Client Dropout. When a client drops out, the other clients in the collaboration no longer get the generalization benefits of accessing a model trained on more diverse data. Recent works [4, 7, 8] have analyzed the effect of client dropout in a cross-device federated learning setting, where the remaining clients still remain connected with the central server. However, in a decentralized learning setting, since there is no central server, the departure of a node will also cause disconnection of its communication links to neighboring nodes. This can disrupt network connectivity, and cause other clients to get forcibly disconnected from the rest of the network. An extreme example is that of a star topology wherein losing the central node isolates all clients, reducing training to fully localized learning. In this paper, we propose a framework to incentivize connectivity-critical clients to continue participating in collaborative decentralized learning to provide generalization benefits to a majority of the clients and also maintain network connectivity.

Example Illustrating the Need to Retain Connectivity-Critical Clients. To illustrate the critical need to retain connectivity-critical clients in decentralized learning, and the effectiveness of our proposed incentivization framework, consider the example in Figure 1 of a network of 12 client nodes. Nodes 3 and 6 have disjoint label distributions compared to nodes 1, 2, 8, 9, 12 and nodes 4, 5, 10, 7, 11. Each client's test set consists of 60% data matching its training distribution and 40% data reflecting the distributions from other clusters. We compare the test accuracies of the clients with the following three different decentralized learning algorithms:

(a) **No Client Dropout Allowed.** The case in Figure 1 (a) corresponds to standard decentralized learning where all clients are required to participate in collaborative training, and implies that all clients must use the single global model $\mathbf{x}$ resulting from the collaboration. Since the model is optimized for the union of the datasets $\cup_i \mathcal{D}_i$, it performs well for most nodes but yields poor test accuracy for nodes 3 and 6, whose local data significantly differ from the global data distribution.

(b) **Client Dropouts Allowed.** For the results illustrated in Figure 1 (b), clients can opt-out and train solo models if collaboration does not benefit them. The low accuracy experienced by nodes 3 and 6 in part (a) causes them to dropout and prefer to train their own models independently. However, the dropout of nodes 3 and 6 also disconnecting the rest of the network causing it to fragment into seven connected components, each of which train their own model. Since each client only learns from its connected peers, reducing its model's exposure to diverse data. This leads to nodes 3 and 6 performing well on their specific label distributions but degrades performance for all other nodes, as they no longer learn from the broader dataset.

(c) **Proposed Incentivized Learning with Client Dropouts Allowed.** To retain nodes 3 and 6, we propose an incentivized learning framework where we modify the objective function so that it penalizes poor accuracy of the shared model on each client's local data. The new objective also gives higher weightage to the retention of connectivity-critical clients, losing whom will disrupt and fragment the rest of the clients in the network. This prevents network fragmentation and results in the shared model performing well of the local datasets of a majority of the clients. In Figure 1 (c) we observe that although other nodes experience a slight drop in performance due to the modification of the objective function, overall, all nodes perform better than in the voluntary dropout scenario, preserving network connectivity and improving global accuracy.

1.1 Our Contributions and Paper Organization

Client participation incentives have only been studied in cross-device federated learning setting with a central aggregating server [4] rather than in a decentralized topology where voluntary dropout of some clients can force disconnection and fragmentation of the network. Thus, to our knowledge, this paper is the first consider the problem of incentivizing clients to participate in decentralized learning, with the purpose of maintaining the shared model's accuracy on clients' local data, while their local accuracy. The paper makes the following key contributions:

- In Section 2, we propose modified objective functions to incentivize key client participation by penalizing their dropouts: one penalizes dropouts of individual clients based on their importance, and other captures the collective importance of group of nodes by penalizing their simultaneous departure (e.g., nodes forming a vertex cut are collectively critical for connectivity).
- In Section 3, we propose two algorithms—`NodeDrop-IDSGD` and `GroupDrop-IDSGD` that adapt client learning rates differently for each client to minimize the proposed dropout-aware objectives.
- In Section 4, we analyze their theoretical convergence and demonstrate a linear speedup with the number of clients in the network.
- In Section 6, we empirically evaluate our methods under flexible client dropouts across various network topologies on image classification tasks. Our results show improved test accuracy and greater retention of nodes in the collaboration network compared

[0]Throughout this work we will use the terms client and node interchangeably

to standard decentralized learning, highlighting the critical importance of retaining key clients and preserving network connectivity for performance.

We omit the proofs and experimental details due to space limitations, and present them in the extended version.

1.2 Related Work

In decentralized learning it is often more critical to retain nodes compared to federated learning since they also form the communication backbone. There are several empirical works that study the impact of losing a fraction of clients on performance in decentralized learning [11, 24, 26]. Beyond loss of data and possible network fragmentation, client dropouts can also affect the convergence rate by altering the network topology. There have been attempts to construct collaboration structure which helps attain faster convergence rates [1, 6, 20]. It has also been empirically demonstrated [25] that in scale-free networks, "central nodes" have a disproportionate impact on the final model performance and convergence rates. Collectively, these studies highlight that certain nodes play a more critical role than others in determining convergence-related metrics in fully decentralized learning. However they do not explicitly try to incentivize retention of such connectivity-critical clients.

Incentivizing client participation has been studied in cross-device federated learning, where a central server aggregates updates from all clients, but not in a decentralized topology. In the centralized federated setting, coalitional game theory has been used to model how self-interested agents form coalitions to maximize local utility ([7, 8]). Several works promote long-term client engagement by incorporating game-theoretic post-training reward mechanisms to incentivize participation ([10, 12, 14, 21, 34, 36]). Some approaches introduce training-time rewards to reduce dropouts [32] during training, while others focus on discouraging dishonest behavior, such as clients submitting large, manipulative updates [3, 9]. The most closely related work to ours changes the global objective function to make the model inherently appealing, encouraging more clients to remain in the collaboration [4]. This approach complements reward-based and personalization methods and can be integrated with them for further incentivization.

2 Problem Formulation

2.1 Notation

For any positive integer N, we define $[N] := \{1, \dots, N\}$. The notation $\|\cdot\|$ or $\|\cdot\|_2$ represents the L_2-norm of a vector. When used for a matrix, $\|\cdot\|_F$ and $\|\cdot\|_2$ denote the Frobenius norm and the spectral norm, respectively. Lowercase boldface letters represent vectors, while uppercase boldface letters represent matrices. The symbol $\mathbb{I}(\cdot)$ denotes the indicator function, and $\mathbf{1}$ represents an all-ones vector of appropriate dimension.

2.2 Decentralized SGD Preliminaries

We consider a scenario where N clients (or nodes) collaborate via a connected network topology to train a shared model parameterized by $\mathbf{x} \in \mathbb{R}^d$. Each client i holds a local dataset $\mathcal{D}_i$, consisting of samples drawn from its respective underlying true data distribution Z_i. The true loss function for client i, given by $f_i(\mathbf{x}) = \mathbb{E}_{\xi \sim Z_i}[l(\mathbf{x}, \xi)]$ and is approximated by the empirical loss $F_i(\mathbf{x}) = \frac{1}{|\mathcal{D}_i|}\sum_{\xi \in \mathcal{D}_i} l(\mathbf{x}, \xi)$, as per the standard empirical risk minimization approach. The N nodes seek to collaboratively minimize the average of the local empirical loss functions, $\sum_{i=1}^{N} F_i(\mathbf{x})/N$, which we sometimes refer to as the global objective function.

Algorithm 1 DECENTRALIZED SGD WITH LOCAL UPDATES

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0,0)} \in \mathbb{R}^d$, stepsizes $\{\eta_t\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

for $t \in 0 \dots T$ **do**
  For each worker $i, i \in [N]$**:**
    For $r = 0, 1 \cdots (\tau - 1)$ **local iterations**
      Sample mini-batch of data $\xi_i^{(t,r)}$,
      Compute local gradient $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
      $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \eta_t \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$; local updates
    $\mathbf{x}_i^{(t+1,0)} := \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$; gossip averaging
end for

The N nodes (or clients) communicate through an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \dots, N\}$ represents the set of nodes, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of undirected edges connecting them. Each edge $(i, j) \in \mathcal{E}$ signifies a bidirectional communication link between nodes i and j. The neighborhood of a node i is given by $\mathcal{N}_i = \{j \mid (i, j) \in \mathcal{E}\}$, representing all nodes directly connected to i.

The standard decentralized SGD algorithm with local updates is given in Algorithm 1. We denote the local model at the starting of round t at node i by $\mathbf{x}_i^{(t,0)}$. Within every round t each client executes τ steps of mini-batch stochastic gradient descent and the intermediate model at node i after $0 \le r \le \tau$ are denoted by $\mathbf{x}_i^{(t,r)}$. We define the vector of these models

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \dots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d \times N}.$$

After local gradient updates at each nodes the local models are updated using weighted average of the neighboring models as $\mathbf{X}^{(t+1,0)} = \mathbf{X}^{(t,\tau)}\mathbf{W}$, where $\mathbf{W} = [w_{ij}]_{i,j \in [N]} \in [0, 1]^{N \times N}$ is referred to the mixing matrix. This averaging step is equivalent to each node performing a weighted average of the neighboring models $\mathbf{x}_i^{(t+1,0)} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$. We assume that the mixing matrix is symmetric ($\mathbf{W} = \mathbf{W}^\top$) and doubly stochastic ($\mathbf{W}\mathbf{1} = \mathbf{1}, \mathbf{1}^\top \mathbf{W} = \mathbf{1}$), satisfies $w_{ij} = 0$ if $(i, j) \notin \mathcal{E}$ and $w_{i,i} \neq 0 \ \forall i \in [N]$. Decentralized SGD relies on the key idea that when the learning rates are sufficiently small, the effect of gossip averaging dominates over local updates, leading to consensus among the models across nodes. The convergence of this algorithm has been analyzed in [16].

2.3 Client Dropout Model

In standard decentralized learning, it is typically assumed that all clients take part in the training process throughout its duration. However, clients whose local datasets differ significantly from the global data distribution often suffer from degraded performance and may want to opt-out of collaborative training if allowed (see Figure 1). Here we will describe a client dropout model that can be used to analyze this behavior.

Each client i seeks a model $\mathbf{x}$ that minimizes its true loss function $f_i(\mathbf{x}) = \mathbb{E}_{\xi\sim Z_i}[l(\mathbf{x},\xi)]$. If $\mathbf{x}_i^{(T,0)}$ denote the model obtained after T rounds, the client has a self-defined threshold ρ_i on its loss $f_i(\mathbf{x}_i^{(T,0)})$ to judge and decide whether to stay in the system. If $f_i(\mathbf{x}_i^{(T,0)}) > \rho_i$ the client decides to dropout and instead choose to train a model on its own in isolation. A good proxy for ρ_i can be obtained by the client i by a few rounds of local training of the model on a held-out dataset. Since the true data distribution Z_i is unknown to the client i, it replaces the true loss $f_i(\mathbf{x}_i^{(T,0)})$ with the empirical loss $F_i(\mathbf{x}_i^{(T,0)})$. Thus if $\mathcal{Y}_i$ as the event that client i drops out,

$$\mathbb{I}(\mathcal{Y}_i) = \mathbb{I}(F_i(\mathbf{x}_i^{(T,0)}) > \rho_i) \tag{1}$$

$$\approx \sigma_i(\mathbf{x}_i^{(T,0)}) \triangleq \frac{1}{1+\exp(-(F_i(\mathbf{x}_i^{(T,0)})-\rho_i))} \tag{2}$$

which is the sigmoid function $F_i(\mathbf{x}_i^{(T,0)}) - \rho_i$. Since the indicator function is non-differentiable, in (2), we approximate it using the sigmoid function, a standard approximation employed in optimization literature.

Remark 1 (Probabilistic interpretation Using the Bradley-Terry Model). *The sigmoid approximation of the indicator admits a probabilistic interpretation, where $\sigma_i(\mathbf{x}_i^{(T,0)})$ can be viewed as the probability that client i drops out, and it aligns with the well-known Bradley-Terry model ([2]) used for pairwise comparisons. As per the Bradley-Terry model, for a score function $r(\mathbf{x})$ indicative of the utility of object $\mathbf{x}$, the probability of preferring an object $\mathbf{x}_a$ over $\mathbf{x}_b$ is expressed as:*

$$\Pr(\mathbf{x}_a \succ \mathbf{x}_b) = \frac{\exp(r(\mathbf{x}_a))}{\exp(r(\mathbf{x}_a)) + \exp(r(\mathbf{x}_b))}.$$

Defining the score for a model as the negative of the loss, $r(\mathbf{x}_i^{(T,0)}) = -F_i(\mathbf{x}_i^{(T,0)})$, and letting ρ_i represent client i's estimate of the achievable loss from local training, giving it a score $r(\mathbf{x}_i^{local}) = -\rho_i$, the probability that client i opts out or prefers the local model over the global model can be expressed as:

$$\Pr(\textit{Client i drops out}) = \Pr(\mathbf{x}_i^{local} \succ \mathbf{x}_i^{(T,0)}) = \frac{\exp(-\rho_i)}{\exp(-\rho_i)+\exp(-F_i(\mathbf{x}_i^{(T,0)}))} = \sigma_i(\mathbf{x}_i^{(T,0)}).$$

This probabilistic interpretation providing a strong theoretical grounding to the proposed dropout model to complement its practical applicability.

2.4 Proposed Modified Objective Functions

Standard decentralized SGD over a connected network topology optimizes the average of the local objective functions, $\sum_{i=1}^N F_i(\mathbf{x})/N$, to obtain a shared model $\mathbf{x}$ that will be later used by the clients in the network. However, due to data heterogeneity, clients whose local data $\mathcal{D}_i$ differs greatly from the union of the datasets $\cup_i \mathcal{D}_i$ will suffer from poor accuracy, as illustrated in Figure 1 part (a). If these dissatisfied clients are allowed to dropout, then it causes accuracy loss for many other clients due to disruption of the connected network topology. We propose the following two ways to modify the objective function: node-wise penalty and group-wise penalty, to encourage connectivity-critical clients to continue participating in the collaboration. Node-wise penalty assigns weights to each node according to that node's local properties, such as its degree, that impact the network's connectivity. Group-wise penalty considers the effect of the departure of groups of nodes on the connectivity.

Modified Objective with Node-wise Penalty. To minimize client dropouts and maximize the number of clients whose performance requirements are met by the resulting model, [4] propose minimizing the objective function given by

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \mathbb{I}(F_i(\mathbf{x}) > \rho_i) \approx \frac{1}{N}\sum_{i=1}^N \sigma_i(\mathbf{x}). \tag{3}$$

This objective function assigns equal importance to each node. However, as discussed earlier, in fully decentralized learning, the importance of nodes is dependent on topology. Thus, it becomes increasingly important to retain key nodes in order to sustain efficient collaboration. To account for this disparity, we propose using a generalized weighting scheme that can incorporate importance weights a_i. We define our modified objective function $\tilde{F}(\mathbf{x})$ as follows

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \tilde{F}_i(\mathbf{x}) := \frac{1}{N}\sum_{i=1}^N a_i\sigma_i(\mathbf{x}) \tag{4}$$

where a_i represents the relative importance of node i. It can be thought of as the penalty for node i leaving the system. For example, one practical choice of a_i is the degree of the node i. From the Bradley-Terry probabilistic interpretation in Remark 1, one can think of $\tilde{F}$ as the expected penalty incurred from dropouts under a model $\mathbf{x}$.

More generally, we can interpolate between minimizing average empirical loss and penalizing key-client dropouts by introducing a mixing parameter $\gamma \in [0,1]$

$$G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N G_i(\mathbf{x}) = (1-\gamma)F(\mathbf{x}) + \gamma\tilde{F}(\mathbf{x}) \tag{5}$$

$$\triangleq (1-\gamma)\left(\frac{1}{N}\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\frac{1}{N}\sum_{i=1}^N a_i\sigma_i(\mathbf{x})\right). \tag{6}$$

When $\gamma = 0$ this reduces to the standard average empirical loss minimization objective. Setting $\gamma = 1$ and $a_i = 1\ \forall i$ recovers the objective function proposed in [4] for incentivizing client participation in the centralized federated learning setting. Notably, [4] shows that incentivizing more clients to participate in training helps the model generalize effectively to unseen inference-time clients who do not participate in collaborative training.

Modified Objective with Group-wise Penalty. In many real-world scenarios, the importance of a node is not solely determined in isolation but rather depends on its relationship with other nodes. Often, groups of nodes collectively contribute to the system's overall stability and function. For example, consider the scenario where nodes a, b, c together form a minimal vertex cut – a critical subset of nodes whose joint removal disconnects the network. If all three nodes were to leave the system, they would fragment the collaboration structure into two disjoint sets, completely severing communication between them. However, if only a subset of these nodes were to leave, the graph would remain connected. In this scenario, we want to associate a penalty with the joint dropout of an entire group of critical nodes. Let $\mathcal{Y}_i$ represent the event that node i leaves the

system. To capture the impact of a complete group departure, we introduce a penalty linked to the event:

$$\mathbb{I}(\mathcal{Y}_a \cap \mathcal{Y}_b \cap \mathcal{Y}_c) = \mathbb{I}(\mathcal{Y}_a)\mathbb{I}(\mathcal{Y}_b)\mathbb{I}(\mathcal{Y}_c) \approx \sigma_a(\mathbf{x}_a^{(T,0)})\sigma_b(\mathbf{x}_b^{(T,0)})\sigma_c(\mathbf{x}_c^{(T,0)})$$

that all three nodes a, b, c simultaneously exit the system, thereby disrupting network connectivity. If only a subset of these nodes leaves the overall network remains connected and thus no penalty is applied. In general, let there be K groups of 'collectively important' nodes where we denote the kth group of nodes by C_k. We define the group-wise penalty function

$$\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{k=1}^{K}\lambda_k \prod_{j\in C_k}\sigma_j(\mathbf{x}) \tag{7}$$

where the λ_k is the weight assigned to group C_k. From the Bradley Terry interpretation in Remark 1, the product $\prod_{j\in C_k}\sigma_j(\mathbf{x})$ can be thought of as the probability of all nodes within the group C_k leaving the system. This is because given the model $\mathbf{x}$, each node independently decides whether to stay or drop out of collaboration, and thus, $\mathcal{Y}_i$ are independent events.

We again define our generalized objective function by interpolating between the empirical risk minimization and group-wise penalty function:

$$G(\mathbf{x}) = (1-\gamma)F(\mathbf{x}) + \gamma\tilde{F}(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^{N}F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^{K}\lambda_k\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)\right\}. \tag{8}$$

Observe that the modified objective function in (8) with the group-wise penalty subsumes the objective in (6) with node-wise penalty by including groups C_k where $|C_k| = 1$ to represent individual nodes.

3 Proposed Incentivized Learning Algorithm

In this section, we propose an incentivized decentralized learning algorithm for the modified objective function in (6). For the modified objective function in (8), the alternative proposed algorithm is in Appendix A due to space limit.

3.1 `NodeDrop-IDSGD` for the Node-wise Penalty Objective

We introduce `NodeDrop-IDSGD`, an algorithm designed to minimize the objective function with node-wise penalty, as defined in Equation (6). Notably, `NodeDrop-IDSGD` does not alter the local model update procedure from Algorithm 1, ensuring that the core optimization framework remains unchanged. At the start of each round $t \in T$, the only additional step in `NodeDrop-IDSGD` is the estimation of an adaptive learning rate for each user (lines 3-6). Once this learning rate is computed, users proceed with the standard τ local stochastic gradient descent (SGD) updates, followed by a gossip averaging step to mix models with their neighbors. Since `NodeDrop-IDSGD` modifies only the learning rate at each user while keeping the rest of the procedure unchanged, it remains a simple yet effective extension of decentralized SGD.

Algorithm 2 `NodeDrop-IDSGD`

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0,0)} \in \mathbb{R}^d$, base LR $\{\eta\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

1: **for** $t \in 0 \dots T$ **do**
2: **For each worker** $i, i \in [N]$**:**
3: **`Learning Rate Estimation Phase:`**
4: Compute $q_i(\mathbf{x}_i^{(t,0)}) = \sigma(\mathbf{x}_i^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)}))$
5: Calculate Pre-factor $b_i^{(t)} = (1-\gamma) + \gamma a_i q_i(\mathbf{x}_i^{(t,0)})$
6: Compute scaling factors $s_i^{(t)}$ using gossip (τ_η steps). `// scalar gossip averaging`
7: Set effective learning rate $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$.
8: **`Model Update Phase:`**
9: **For** $r = 0, 1 \cdots (\tau-1)$ **local iterations**
10: Sample $\xi_i^{(t,r)}$,
11: Compute $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
12: $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
13: $\mathbf{x}_i^{(t+1,0)} := \sum_{j\in N_i} w_{ij}\mathbf{x}_j^{(t,\tau)}$ `// model gossip averaging`
14: **end for**

The gradient of the modified objective function G is a weighted average of the gradients of the empirical loss functions F_i given by

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N}\left((1-\gamma) + \gamma a_i \underbrace{\sigma(\mathbf{x})(1-\sigma(\mathbf{x}))}_{q_i(\mathbf{x})}\right)\nabla F_i(\mathbf{x}).$$

We denote the pre-factor that is multiplied to the gradient $\nabla F_i(\mathbf{x})$ at node i at time t by

$$b_i^{(t)} = (1-\gamma) + \gamma a_i q_i(\mathbf{x}_i^{(t,0)})$$

where, according to our client dropout model, $q_i(\mathbf{x})$ takes a larger value if the client is dissatisfied with model $\mathbf{x}$ and its dropout probability is close to 0.5.

If we were to apply simple gradient descent to minimize G_i, the update rule would be given by $\mathbf{x}_i^{(t+1)} \leftarrow \mathbf{x}_i^{(t)} - \eta\nabla G_i(\mathbf{x}_i^{(t)}) = \mathbf{x}_i^{(t)} - \eta b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t)})$. Here, η can be interpreted as the learning rate with respect to G_i, while $\eta b_i^{(t)}$ serves as the effective learning rate with respect to F. Multiplication by the pre-factor can be used to account for the relation between the gradients of G and F.

Learning Rate Estimation Phase: The process starts with a simple inference step (line 4) wherein each client computes $q_i(\mathbf{x}_i^{(t,0)}) = \sigma(\mathbf{x}_i^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)}))$. In Algorithm 2, $\hat{\eta}_i^{(t)}$ can be interpreted as the learning rate with respect to F, while $\frac{\eta}{\tau(s_i^{(t)}+\epsilon)}$ corresponds to the learning rate with respect to G. The normalization by τ accounts for the number of local updates, is simply a matter of convention and does not affect our results. The term $s_i^{(t)}$ denotes the scaling factor at client i used to approximate the global sum of pre-factors $s^{(t)} := \sum_{i=1}^{N} b_i^{(t)}$ for normalization. This results in the final learning rate at each client: $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$. The smoothness parameter of the global objective function's (G) landscape ($\tilde{L}_s$) depends on the point

$\mathbf{x}$ and takes the form $A + B\sum_{i=1}^{N}((1-\gamma) + \gamma a_i q_i(\mathbf{x}))$, as shown in Lemma 2 (Appendix C). Following standard practice in stochastic optimization, we adapt the learning rate with respect to G to be $\frac{1}{\tilde{L}_s} = \frac{1}{A+B\sum_{i=1}^{N} b_i^{(t)}}$ This dynamic normalization is incorporated into our algorithm by setting the learning rate for G to $\frac{\eta}{\epsilon+\sum_{i=1}^{N} b_i^{(t)}}$, where η and ϵ are tunable hyperparameters, and the denominator acts as a dynamic normalization factor. Lastly note that we compute this learning rate estimation procedure only once for every τ local SGD steps.

Estimating Scaling Factors: The scaling factors $s_i^{(t)}$ are estimated using gossip averaging with the same mixing matrix $\mathbf{W}$. Each user i only has access to its local pre-factor $b_i^{(t)}$ and direct communication between all nodes is impractical in a decentralized setup. To address this we use gossip averaging, which enables nodes to iteratively exchange information with their neighbors to collaboratively approximate $s^{(t)}$. To estimate $s^{(t)}$, each user participates in τ_η steps of gossip averaging, initializing the process with $s_i^{(t,0)} = Nb_i^{(t)}$. This ensures that each node starts with a scaled version of its local term before iteratively refining its estimate. During gossip averaging, the scaling factors are iteratively updated using the rule: $s_i^{(t,r_\eta+1)} = \sum_{j\in\mathcal{N}_i} w_{ij} s_j^{(t,r_\eta)}$ which can be expressed in matrix form as: $\mathbf{s}^{(t,r_\eta+1)} = \mathbf{s}^{(t,r_\eta)}\mathbf{W}$, with $\mathbf{s}^{(t,r_\eta)} = [s_1^{(t,r_\eta)} \cdots s_N^{(t,r_\eta)}]$ for $0 \le r_\eta \le \tau_\eta - 1$. After τ_η steps of gossip averaging node i estimates $s^{(t)}$ as

$$s_i^{(t)} = \max\left\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\right\} = \max\left\{s_i^{(t,\tau_\eta)}, b_i^{(t)}\right\}.$$

This step ensures that the estimate remains robust and non-degenerate, preventing cases where averaging might lead to values that are too small. By enforcing a lower bound of $b_i^{(t)}$ we guarantee that the estimate of the sum the pre-factors of all nodes $\in [N]$ at node i is larger than the pre-factor at node i ($b_i^{(t)}$). Thus finally client i approximates the effective learning rate $\left(\frac{\eta b_i^{(t)}}{\tau(s^{(t)}+\epsilon)}\right)$ using $\hat{\eta}_i^{(t)} = \frac{\eta b_i^{(t)}}{\tau(s_i^{(t)}+\epsilon)}$.

REMARK 2 (MINIMAL COMMUNICATION OVERHEAD FOR SCALING FACTOR ESTIMATION). *The τ_η gossip averaging steps used to estimate the scaling factor. Since the scaling factor is a one-dimensional quantity it incurs a negligible communication cost compared to the model averaging step, which involves exchanges of high-dimensional deep neural network parameters (line 12 of Algorithm 2).*

Model Update Phase: Given the modified learning rate the model update proceeds in the same way as decentralized SGD (Algorithm 1). For the rth local step in round t each client i independently samples a mini-batch $\xi_i^{(t,r)}$, computes the gradient w.r.t. $\xi_i^{(t,r)}$ and updates the model parameters.

4 Theoretical Guarantees

In this section we will discuss the convergence guarantees for our proposed algorithms. We start by listing some assumptions commonly used in Federated, Decentralized learning and optimization literature. [15–17, 27, 28]. Note that we will not make any assumptions about our modified objective functions; rather, all our assumptions are limited to the local empirical loss functions F_i.

ASSUMPTION 1. *(Unbiased Stochastic Gradient with Bounded Variance) : For any mini batch ξ_i sampled uniformly from the dataset we have the expected value of the stochastic gradient $\mathbb{E}[\nabla F_i(\mathbf{x}, \xi_i)] = \nabla F_i(\mathbf{x})$ and its variance bounded as $\mathbb{E}[\|\nabla F_k(\mathbf{x}_k, \xi_k) - \nabla F_k(\mathbf{x}_k)\|^2] \le \sigma^2$.*

ASSUMPTION 2. *(Continuity & Smoothness of local objective functions) : The local objective functions $F_1(\mathbf{x}), \ldots, F_N(\mathbf{x})$, are L_c-continuous and L_s-smooth for any $\mathbf{x}$.*

ASSUMPTION 3. *(Bounded Dissimilarity of $F(\mathbf{x})$) : There exist constants κ and β such that*

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla F_i(\mathbf{x})\|_2^2 \le \kappa^2 + \beta^2\|\frac{1}{N}\sum_{i=1}^{N}\nabla F(\mathbf{x})\|_2^2.$$

ASSUMPTION 4. *(Mixing matrix and Consensus Rate) We assume that the mixing matrix is symmetric ($\mathbf{W} = \mathbf{W}^\top$) and doubly stochastic ($\mathbf{W1} = \mathbf{1}, \mathbf{1}^\top\mathbf{W} = \mathbf{1}$), satisfies $w_{ij} = 0$ if $(i,j) \notin \mathcal{E}$ and $w_{i,i} \neq 0$ $\forall i \in [N]$. In addition, we assume that there exists a constant $p \in (0,1]$ such that for any $\mathbf{X} \in \mathbb{R}^{d\times N}$ we have $\|\mathbf{XW} - \bar{\mathbf{X}}\|_F^2 \le (1-p)\|\mathbf{X} - \bar{\mathbf{X}}\|_F^2$. Here $\bar{\mathbf{X}} = \mathbf{X}\frac{\mathbf{11}^T}{N}$.*

This property always holds for p equal to the second largest eigenvalue of the matrix W given by $\left\|\mathbf{W} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right\|_2$. A larger p means better connectivity and more efficient information mixing, p for a fully connected/federated topology is 1. The convergence is typically analyzed in terms of an average virtual model at the start of round t, denoted by

$$\bar{\mathbf{x}}^{(t,0)} := \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i^{(t,0)}.$$

THEOREM 1 (CONVERGENCE OF NODEDROP-IDSGD). *For the objective function G with nodewise penalty defined in equation (6), under assumptions 1-4 there exists a base learning rate η under which for Algorithm 2 we have*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\bar{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \le \mathcal{O}\left(\sqrt{\frac{G_{\text{diff}}(1+\epsilon_\eta)}{N\tau T}}\right)$$
$$+ \mathcal{O}\left(\frac{\left(G_{\text{diff}}\right)^{2/3}(1+\epsilon_\eta^{1/3})}{p^{2/3}T^{2/3}}\right) + \mathcal{O}\left(\frac{\left(G_{\text{diff}}\right)^{2/3}(1+\epsilon_\eta^{1/3})}{p^{2/3}\tau^{1/3}T^{2/3}}\right)$$
$$+ \mathcal{O}\left(\frac{\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{G_{\text{diff}}}{T}\right) + \mathcal{O}\left(\epsilon_\eta\right).$$

Here $G_{\text{diff}} = G(\bar{\mathbf{x}}^{(0,0)}) - G_{\text{inf}}$, is the initial distance from the infimum and $\mathcal{O}$ subsumes all constants. The parameter τ denotes the number of local steps performed in each iteration. The term $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ represents the learning rate estimation error, where τ_η is the number of gossip averaging steps used for estimating the scaling factors. The initial consensus error defined as $\Xi_0^2 = \frac{1}{N}\sum_{i=1}^{N}\|\bar{\mathbf{x}}^{(0,0)} - \mathbf{x}_i^{(0,0)}\|^2$.

The proof is deferred to Appendix C. The only term that does not vanish as $T \to \infty$ is the learning rate estimation error $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$. This term, however, can be made arbitrarily small by increasing τ_η, independent of the other terms with negligible communication overhead (see Remark 2). We observe a linear speedup in both the

number of clients N and the number of local steps τ for the dominant term ($O(T^{-1/2})$), which decays the slowest with T.

5 Measures of Node Importance

In NodeDrop-IDSGD, the importance weights a_i control how strongly the algorithm penalizes the dropout of client i. Clients with larger a_i incur a higher penalty when they leave, translating into larger effective learning rates and a stronger incentive to remain in the collaboration. The choice of a_i is therefore the mechanism by which we encode which clients are most critical for accuracy and connectivity.

We consider a broad family of weighting rules $a = (a_1, \ldots, a_N)$, all implemented through a common interface that takes as input the communication graph G. The strategies we evaluate fall into three classes: topology-based, data-based, and hybrid.

5.1 Importance Derived from Topology

Topology-based strategies assign importance using only the structure of the communication graph. We consider the following choices:

Degree. Assigns larger weights to nodes with more neighbors that provide alternative paths and accelerate gossip mixing; this choice is easy to compute and aligns with intuitive connectivity, but reflects only local structure.

Degree (normalized). Uses a normalized version of degree so that the average (or maximum) weight is close to one, making weights more comparable across graphs of different sizes and densities; it preserves the intuition of Degree while avoiding excessively large a_i on dense topologies, but still ignores global structure beyond immediate neighbors.

Betweenness. Sets a_i proportional to how often node i lies on shortest paths between other nodes, thereby prioritizing "bridge" clients that mediate many communication routes; this protects connectivity bottlenecks, but can be sensitive to small perturbations of the graph.

Eigenvector. Assigns larger a_i to clients that are connected to other highly weighted clients, capturing influence in dense core regions rather than just local degree; this is well suited to emphasizing hubs in well- connected subgraphs, but may over-emphasize nodes inside tightly knit clusters.

Closeness. Sets a_i according to how close node i is (in shortest-path distance) to all other nodes, favoring clients that can quickly reach the rest of the network; this can accelerate information propagation, but is less focused on explicit cut vertices.

PageRank. Defines a_i via a random-walk-based importance score, assigning larger weights to nodes that are visited frequently by a random walk on G; this measure is robust and relatively easy to compute, but its notion of importance is more diffuse and does not always coincide with strict connectivity criticality.

Vertex connectivity. Assigns larger a_i to nodes whose removal causes the largest drop in global vertex connectivity or markedly increases the number or imbalance of connected components; it directly targets overall network robustness, but is computationally more demanding and may be unnecessary for small or very dense graphs.

5.2 Importance Derived from Data Distribution

Data-based strategies ignore the graph structure and derive a_i from the empirical label distribution p_i at client i, capturing how atypical or specialized each client's data are:

KL Divergence. Sets $a_i = 1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, where p_{global} is the average label distribution across clients and $\mathrm{KL}(\cdot\|\cdot)$ is the Kullback–Leibler divergence [19] (a non-negative measure of how different two discrete distributions are). This assigns larger weights to clients whose data deviate most from the global distribution, thereby prioritizing sources of rare or non-representative data, but can overemphasize very small yet highly idiosyncratic datasets.

Inverse Entropy. Sets $a_i = 1 + (H_{\max} - H(p_i))/H_{\max}$, where $H(p_i)$ is the entropy of the local label distribution and $H_{\max}$ its maximum value; clients with low entropy (highly skewed data) receive larger weights, reflecting both their higher dropout risk and their role in representing rare modes, but clients with nearly uniform data are treated as less critical even if they are numerous.

Dropout Risk (DR). Combines divergence and entropy into a single score $a_i = 1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})\big(1 - H(p_i)/H_{\max}\big)$, so that clients are emphasized when they are both far from the global distribution and highly specialized; this targets exactly the nodes we expect to be most likely to leave and most costly to lose, at the expense of slightly more computation and sensitivity to how divergence and specialization are balanced.

REMARK 3. *In these data-imbalance scenarios, our modified objective functions can also be applied in a traditional (centralized) federated learning setting. When a central server manages aggregation, the same importance weights a_i can be used without incurring the estimation errors that arise in a fully decentralized coordination.*

5.3 Hybrid Importance Measures

Hybrid strategies combine structural and statistical signals, assigning higher importance to clients that are simultaneously critical for connectivity and for representing heterogeneous data:

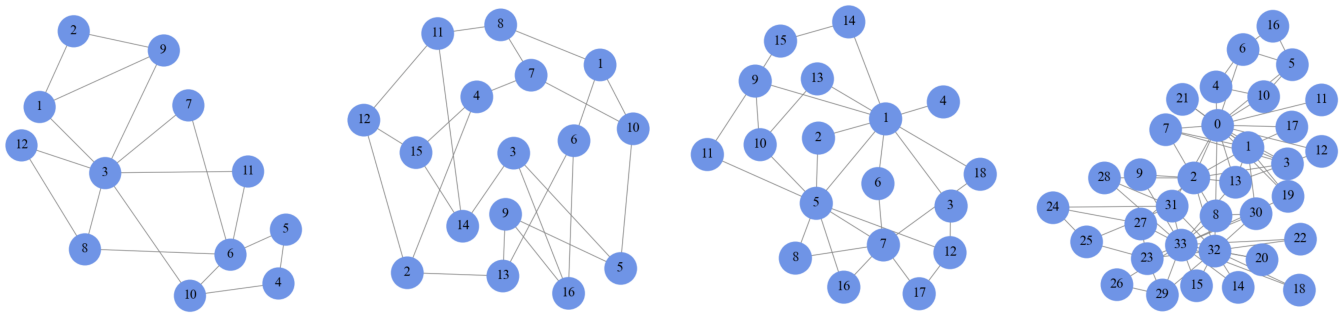
Degree with KL. Sets $a_i = \deg(i)\big(1+\mathrm{KL}(p_i \,\|\, p_{\text{global}})\big)$, emphasizing clients that are both highly connected and whose label distribution differs substantially from the global one; this captures hubs with atypical data, but can overweight nodes in dense regions of the graph.

Betweenness with KL. Scales betweenness-centrality weights by $1+\mathrm{KL}(p_i \,\|\, p_{\text{global}})$, so that clients lying on many shortest paths and also having heterogeneous data receive the largest a_i; this protects bridge nodes whose dropout would both fragment the graph and remove access to rare data modes, but inherits betweenness centrality's sensitivity to small changes in the topology.

Betweenness with DR. Multiplies betweenness-based weights by the Dropout Risk score, thus prioritizing bridges that are also highly specialized and far from the global distribution; this directly targets clients that are both structurally and statistically fragile, at the cost of slightly higher computational overhead.

Eigenvector with KL. Combines eigenvector centrality strategie with $1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$ to emphasize clients that are embedded in influential core regions of the graph and whose data are atypical; this is well suited to capturing important hubs with rare labels, but may focus on dense clusters more than on sparsely connected bridges.

PageRank with KL. Weights clients according to a random-walk-based importance score scaled by $1 + \mathrm{KL}(p_i \,\|\, p_{\text{global}})$, highlighting



(a) Scale Free Like 12 (b) 16 Node 3 Connections (c) Scale Free Like 18 (d) Karate Club Network

Figure 2: Graph Topologies considered in our experiments

nodes that are frequently visited by random walks and also carry heterogeneous data; this yields a robust and relatively smooth importance profile, though its notion of centrality is more diffuse than strict cut-based measures.

6 Experiments

All the experiments are done on the following graph topologies, illustrated in Figure 2.

(1) **Scale Free Like 12 (SFL12)** - A network of 12 clients with node degrees sampled to roughly follow a power-law distribution, mimicking the structure of a scale-free network. In such networks, a few nodes act as hubs with high connectivity, while most have low degrees—reflecting heterogeneity often seen in real-world systems like social or communication networks. Nodes 3 and 6 here are very critical to connectivity.
(2) **16 Node 3 Connections (3Con16)** A uniform network of 16 clients where each node has exactly 3 connections (degree 3). This regular structure implies equal importance when node degree is used as a centrality or importance metric. The network is harder to fully disconnect due to its regular and balanced connectivity.
(3) **Scale Free Like 18 (SFL18)** - An 18-client extension of the SFL 12 network, preserving the general shape of a power-law-inspired degree distribution.
(4) **Karate Club Network (KCN)** - A well-known 34-node social network ([35]) with a few high degree nodes, It features a skewed degree distribution with an average degree of 4.59, a maximum of 17, and a minimum of 1.

The mixing weights are chosen according to Metropolis-Hastings mixing matrix construction where $w_{i,j} = \frac{1}{1+\max\{\deg(i),\deg(j)\}}$ for $i \neq j$, $(i,j) \in \mathcal{E}$ and $w_{i,i} = 1 - \sum_{j \in \mathcal{N}_i} w_{i,j}$. Note that this gives us a symmetric and doubly stochastic matrix $\mathbf{W}$.

6.1 Experimental Setup

To evaluate the effectiveness of our methods, we consider a dynamic setting where clients are allowed to flexibly opt out of the system after 30% of the training rounds. When a client drops out, all of its connections to other nodes in the network are severed. This setup highlights the critical role of high-degree nodes: their departure not only renders their local data inaccessible but also risks fragmenting the network if multiple such nodes exit simultaneously. Clients that leave the collaborative training continue with solo training on their local datasets, while the remaining clients continue to train collaboratively within the subnetwork that remains connected to them after the dropouts using a new mixing matrix calculated using metropolis hastings, benefiting only from the data available within that connected subnetwork.

Dataset, Models and Metrics: We evaluate the performance of our proposed methods on the following image classification datasets i) FashionMNIST ([33]) with 10 labels, ii) EMNIST-Balanced ([5]) with 47 labels and iii) CIFAR-10 ([18]) with 10 labels.

We evaluate our method's performance against standard decentralized learning using preferred model test accuracy, defined as the average test accuracy across clients obtained from the collaboratively trained model for clients that remain in the system, and from the solo-trained model for clients that drop out of collaboration.

6.2 Experimental Results

Table 1 reports the preferred test accuracy and retention rate for decentralized SGD (with and without dropouts) and NodeDrop-IDSGD across four datasets and four topologies. On Fashion-MNIST, NodeDrop-IDSGD consistently recovers the loss due to voluntary dropouts and even exceeds the no-dropout baseline on all graphs (for instance, 69.0% to 77.1% on SFL12, above the 75.2% baseline). On EMNIST, improvements are larger: accuracy roughly doubles on 3Con16 and increases by more than 20 percentage points on SFL12 and SFL18, while on KCN it moves from the mid-30% to the high-40% range, still below the no-dropout baseline but substantially better than with dropouts. On CIFAR-10, accuracy rises from the low 30% range to the high 40% range across topologies. On CelebA, NodeDrop-IDSGD improves over the dropout baseline by about 10–14 percentage points on SFL12, 3Con16, and SFL18, with a smaller but positive gain on KCN. Retention exhibits the same trend. On KCN, retention also improves, reaching about 20% on CelebA and more than 75% on EMNIST. On the more regular 3Con16 graph, retention gains are modest, reflecting the higher inherent robustness of this topology.

Table 2 compares the accuracy of NodeDrop-IDSGD under different importance-weight strategies a_i across datasets and topologies.

		Fashion MNIST				EMNIST				CIFAR-10				CelebA			
		Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ	Base	Drop.	**Ours**	Δ
SFL12	Accuracy	75.2	69.0	77.1	+8.10	57.1	46.8	70.5	+23.7	36.8	31.4	48.1	+16.7	61.7	61.1	72.4	+11.3
	Ret. Rate	N/A	44.2	81.2	+37.0	N/A	73.3	100	+26.7	N/A	0.0	21.7	+21.7	N/A	4.2	41.7	+37.5
3Con16	Accuracy	74.1	66.6	74.7	+8.10	54.1	36.3	69.2	**+32.9**	31.9	25.3	42.8	+17.5	58.7	57.0	70.8	+13.8
	Ret. Rate	N/A	37.5	64.4	+26.9	N/A	51.3	99.4	**+48.1**	N/A	0.00	11.3	+11.3	N/A	1.30	23.8	+22.5
SFL18	Accuracy	74.9	66.1	78.4	+12.3	57.2	43.6	66.2	+22.6	37.3	32.5	48.3	+15.8	60.7	60.3	70.4	+10.1
	Ret. Rate	N/A	41.7	65.0	+23.3	N/A	66.3	98.9	+32.6	N/A	0.00	27.8	+27.8	N/A	8.90	30.6	+21.7
KCN	Accuracy	77.0	68.2	69.1	+0.90	59.6	34.3	49.4	+15.1	41.9	37.4	48.1	+10.7	65.9	65.6	66.4	+0.80
	Ret. Rate	N/A	55.9	55.9	+0.00	N/A	50.0	75.9	+25.9	N/A	0.00	36.2	+36.2	N/A	3.80	20.3	+16.5

Table 1: Performance of Decentralized Learning (baseline or with dropouts) and Incentivized Learning under different network topologies. Table shows the average preferred model's test accuracy (solo trained/from collaboration) and percentage of clients retained in the largest connected component of the network after client dropouts.

On Fashion-MNIST, the hybrid *Degree with KL* strategy gives the best accuracy on SFL12, while *Vertex connectivity* is best on SFL18 and the simple *Closeness* rule is best on KCN. For EMNIST, *Degree with KL* again performs best on SFL12 and SFL18, whereas *Closeness* and *PageRank* are best on 3Con16 and KCN. On CIFAR-10, *Betweenness* is best on SFL12, *Betweenness with KL* on 3Con16, and the hybrids *Degree with KL*, *Eigenvector with KL*, and *PageRankwith KL* achieve the highest accuracies on SFL18 and KCN.

Discussion. Results from Table 1 show that NodeDrop-IDSGD almost always matches or exceeds the preferred accuracy of the no-dropout baseline (the only exception being EMNIST on KCN), while substantially improving over the dropout baseline across all settings. The largest benefits appear on heterogeneous, hub-dominated topologies (SFL12, SFL18, KCN), where explicitly penalizing client dropouts helps avoid fragmentation of the collaboration graph and leads to significant accuracy gains.

Table 2 results indicate that no single weighting rule dominates across all datasets and topologies, but a few clear patterns emerge. Hybrid strategies that combine structural and statistical information tend to perform best on highly heterogeneous topologies (SFL12, SFL18, KCN), especially on EMNIST and CIFAR-10. Purely topology-based centrality measures (*Closeness*, *Vertex connectivity*, and *Betweenness*) also achieve strong performance, particularly on Fashion-MNIST and simpler graphs, while data-only strategies like *KL Divergence* can be most effective when label heterogeneity is pronounced, as in CelebA. Together, these trends suggest that incorporating both connectivity and data heterogeneity into a_i is most beneficial in fragile networks, whereas simpler topology-based weights already work well in more regular settings.

7 Concluding Remarks

In this paper, we investigate the impact of client dropouts on the overall performance of decentralized learning systems and explore incentive mechanisms to mitigate these effects. We propose a generalized framework that incorporates dropout penalties directly into the objective function, enabling the system to discourage early exits. Our proposed algorithm, supported by convergence guarantees, show that the framework effectively incentivizes client participation, leading to improved client retention (up to +48%) within the network and, consequently, better generalization and higher test accuracy, with improvements of up to 33% in our experiments.

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	Fashion MNIST				EMNIST				CIFAR-10				CelebA			
Strategy (a_i)	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN	SFL12	3Con16	SFL18	KCN
Topology-based																
Degree	73.1	74.4	59.1	58.1	69.6	68.3	59.9	40.5	43.8	42.3	47.1	43.2	68.3	70.0	69.4	65.7
Degree (norm.)	73.7	73.6	58.6	65.3	67.8	66.7	58.7	46.2	43.3	41.8	45.6	44.0	69.6	69.8	69.5	65.8
Betweenness	67.9	74.4	53.1	68.3	68.2	68.2	63.5	40.8	**48.1**	42.3	44.9	46.8	63.2	70.0	62.7	64.2
Eigenvector	66.9	73.5	63.2	54.7	67.3	66.7	55.0	33.1	43.3	41.8	44.8	38.9	69.6	69.8	69.8	64.6
Closeness	76.8	70.8	62.2	58.2	70.2	**69.2**	65.6	33.8	43.2	42.5	45.4	40.7	70.3	70.0	69.3	63.8
PageRank	73.4	73.5	57.3	62.0	67.8	66.7	58.8	42.1	43.2	41.8	45.6	44.1	69.8	69.8	69.7	**66.4**
Vertex connect.	70.6	74.2	**70.0**	**69.1**	68.3	68.1	59.0	30.9	42.7	42.0	45.8	45.5	69.7	69.9	67.8	62.8
Data-based																
KL Divergence	72.3	74.7	61.8	60.4	69.3	68.2	56.6	40.5	42.7	42.1	43.9	44.9	**72.4**	70.6	**70.4**	62.6
Inverse Entropy	69.2	73.7	64.3	61.3	68.3	67.4	55.7	31.5	43.3	41.9	45.1	42.7	69.3	68.2	69.6	62.6
Dropout Risk	75.9	70.5	54.6	64.1	68.4	67.5	55.9	30.2	43.4	42.1	43.6	44.6	69.8	70.7	69.1	62.6
Hybrid																
Degree_x_KL	**77.1**	71.1	66.8	62.0	**70.5**	69.2	**66.2**	47.1	45.3	42.8	47.7	**48.1**	68.6	70.8	68.0	65.1
Betweenness_x_KL	69.9	71.1	60.7	61.2	68.6	69.1	63.1	41.8	46.0	**42.8**	45.3	46.7	63.1	**70.8**	56.7	63.4
Betweenness_x_DR	68.5	70.9	59.3	62.5	68.3	68.8	63.0	41.7	46.2	42.5	45.3	46.7	63.1	70.7	56.8	63.1
Eigenvector_x_KL	73.4	**74.7**	63.5	60.1	69.3	68.2	65.0	36.1	44.6	42.1	48.1	38.6	67.7	70.6	69.4	63.2
PageRank_x_KL	76.9	74.7	62.1	63.9	69.6	68.2	59.5	**49.4**	45.0	42.1	**48.3**	44.4	67.8	70.6	68.8	64.6

Table 2: Test accuracy (%) of NodeDrop-IDSGD under different importance weight strategies a_i across four datasets and four network topologies. Each test was run with 10 different seeds.

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A Proposed Incentivized Learning Algorithm (Alternative)

We also propose an incentivized decentralized learning algorithm for the modified objective function in (8). Similar to Algorithm 2, it maintains the local model update procedure at the clients unchanged only adaptively adjusting the learning rate at the beginning of each round (lines 3-8) to perform this minimization.

A.1 Modified Objective with group-wise Penalty: `GroupDrop-IDSGD`

Algorithm 3 `GroupDrop-IDSGD`

Require: For each node $i \in [N]$, initialize $\mathbf{x}_i^{(0)} \in \mathbb{R}^d$, base LR $\{\eta\}_{t=0}^{T-1}$, no. of iterations T, mixing matrix $\mathbf{W}$

1: **for** $t \in 0 \dots T$ **do**
2: **For each worker** $i, i \in [N]$**:**
3: **Learning Rate Estimation Phase:**
4: Compute $\sigma(\mathbf{x}_i^{(t,0)})$
5: Compute M_{ki} $\forall k \in [K]$, $i \in [N]$ using log gossip (τ_b steps). `// low dim vector averaging`
6: Estimate $\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k \in \mathcal{K}_i} \lambda_k M_{ki}^{(t)} (1 - \sigma(\mathbf{x}_i^{(t,0)}))$
7: Compute scaling factors $s_i^{(t)}$ using gossip averaging (τ_η steps). `// scalar averaging`
8: Set the effective learning rate $\hat{\eta}_i^{(t)} = \frac{\eta \hat{b}_i^{(t)}}{\tau(s_i^{(t)} + \epsilon)}$.
9: **Model Update Phase:**
10: **For** $r = 0, 1 \cdots (\tau - 1)$ **local iterations**
11: Sample $\xi_i^{(t,r)}$,
12: Compute $\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$
13: $\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})$ `// stochastic gradient updates`
14: $\mathbf{x}_i^{(t+1,0)} := \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{(t,\tau)}$ `// model gossip averaging`
15: **end for**

The gradient of the modified global objective function (equation (8)) is given by

$$\nabla G(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} (1-\gamma) \nabla F_i(\mathbf{x}) + \frac{1}{N} \gamma \sum_{k=1}^{K} \sum_{l \in C_k} \lambda_k \left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right) (1 - \sigma_l(\mathbf{x})) \nabla F_l(\mathbf{x}). \tag{9}$$

Let $\mathcal{K}_i := \{k | i \in C_k\} \subseteq [K]$ be the set of all groups which node i is a part of. One can rewrite the gradient of the modified objective in equation (9) as

$$\nabla G(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} (1-\gamma) \nabla F_i(\mathbf{x}) + \frac{1}{N} \sum_{i=1}^{N} \left(\gamma \sum_{k \in \mathcal{K}_i} \lambda_k \left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right) (1 - \sigma_i(\mathbf{x})) \right) \nabla F_i(\mathbf{x}).$$

Learning rate estimation phase: Note that this gradient can be expressed as a summation of terms over i. Every object in the ith term is computable at node i, except for $\left(\prod_{j \in C_k} \sigma_j(\mathbf{x}) \right)$. This product for $k \in \mathcal{K}_i$ must be estimated at node i in order to compute the gradient of the global objective function using the existing gossip infrastructure. Let $M_k^{(t)} := \prod_{j \in C_k} \sigma_j(\mathbf{x}_j^{(t,0)})$ denote the product associated with the kth group, and let $M_{ki}^{(t)}$ represent its estimate at node i at the beginning of round t. Note that this estimation is carried out only once every τ local SGD steps at each node. Once the estimation is completed, node i approximates the pre-factor $b_i^{(t)}$ at round t (line 6) as

$$\hat{b}_i^{(t)} := (1-\gamma) + \gamma \sum_{k \in \mathcal{K}_i} \lambda_k M_{ki}^{(t)} (1 - \sigma_i(\mathbf{x}_i^{(t,0)})).$$

Estimating Pre-factors: To estimate $M_k^{(t)} := \prod_{j \in C_k} \sigma_j(\mathbf{x}_j^{(t,0)})$, we use τ_b steps of log gossip. The idea of log gossip is simple – given a method that computes the average of N terms, one can estimate the product of N terms $[d_1, d_2, \dots, d_N]$ by first taking their logarithms and scaling by N to get $[N \ln(d_1), N \ln(d_2), \dots, N \ln(d_N)]$, averaging these via gossip over the network, and then exponentiating the result to

recover $\prod_i d_i$. Since the decentralized network topology serves as the communication backbone, every node participates in the log gossip procedure for all K groups. Define $\mathbf{m}_i^{(t,r_b)} := \left[M_{1i}^{(t,r_b)}, M_{2i}^{(t,r_b)} \cdots M_{Ki}^{(t,r_b)}\right]^\top \in \mathbb{R}^K$ as the vector of estimates at node i during the averaging step r_b at round t where $0 \le r_b \le \tau_b - 1$. Define the point wise log of this vector by

$$\begin{aligned}\mathbf{v}_i^{(t,r_b)} &\triangleq \ln\left(\mathbf{m}_i^{(t,r_b)}\right)\\ &= \left[\ln\left(M_{1i}^{(t,r_b)}\right), \ln\left(M_{2i}^{(t,r_b)}\right) \cdots \ln\left(M_{Ki}^{(t,r_b)}\right)\right]^\top\\ &= \left[v_{1i}^{(t,r_b)}, v_{2i}^{(t,r_b)} \cdots v_{Ki}^{(t,r_b)}\right]^\top.\end{aligned}$$

We perform gossip averaging on this vector. Specifically, we initialize

$$v_{ki}^{(t,0)} = \begin{cases} N \ln\left(\sigma_i(\mathbf{x}_i^{(t,0)})\right), & \text{if } k \in \mathcal{K}_i,\\ 0, & \text{otherwise.}\end{cases}$$

The update rule for gossip averaging follows the standard consensus form given by $\mathbf{v}_i^{(t,r_b+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\mathbf{v}_i^{(t,r_b)}$ or equivalently for the matrix $\mathbf{V}^{(t,r_b)} = \left[\mathbf{v}_1^{(t,r_b)}, \mathbf{v}_2^{(t,r_b)} \cdots \mathbf{v}_N^{(t,r_b)}\right] \in \mathbb{R}^{N\times K}$ the update rule in matrix form is given by $\mathbf{V}^{(t,r_b+1)} = \mathbf{V}^{(t,r_b)}\mathbf{W}$. After τ_b steps of log gossip we compute the final estimate $M_{ki}^{(t)}$ as

$$M_{ki}^{(t)} = \min\left\{M_{ki}^{(t,\tau_b)}, \left(M_{ki}^{(t,0)}\right)^{\frac{1}{N}}\right\}.$$

Where $M_{ki}^{(t,\tau_b)} = \exp(v_{ki}^{(t,\tau_b)})$ and $M_{ki}^{(t,0)} = \exp(v_{ki}^{(t,0)})$. Which is equivalent to

$$M_{ki}^{(t,\tau_b)} = \begin{cases} \min\left\{M_{ki}^{(t,\tau_b)}, \sigma_i(\mathbf{x}_i^{(t,0)})\right\} & k \in \mathcal{K}_i\\ \min\left\{M_{ki}^{(t,\tau_b)}, 1\right\} = M_{ki}^{(t,\tau_b)} & k \notin \mathcal{K}_i\end{cases}$$

The minimization is performed to ensure non-degenerate estimates, wherein, when $k \in \mathcal{K}_i$, the estimate $M_{k,i}^{(t)}$ of $M_k^{(t)} := \prod_{j\in\mathcal{C}_k} \sigma_j(\mathbf{x}_j^{(t,0)})$ does not exceed $\sigma_i(\mathbf{x}_i^{(t,0)})$ (since $\sigma_i(\cdot) \le 1$), thereby ensuring a reasonable approximation. Lastly, note that these τ_b steps of averaging are performed over vectors of dimension K, resulting in a negligible communication cost compared to the model averaging step, where nodes exchange high-dimensional models, which in the case of deep neural networks can potentially have hundreds of millions of parameters.

REMARK 4 (MINIMAL COMMUNICATION OVERHEAD FOR PRE-FACTOR AND SCALING FACTOR ESTIMATION). *The τ_η and τ_b steps of gossip averaging used for estimating the scaling factor (a one-dimensional quantity) and the pre-factor (products M_{ki} estimated jointly as K-dimensional vector) respectively, incur negligible communication cost as compared to the model averaging step, which involves exchanges of high-dimensional deep neural network parameters (line 14 on Algorithm 3).*

THEOREM 2 (CONVERGENCE OF GROUPDROP-IDGSD). *For the objective function G with groupwise penalty defined in equation* (8)*, under assumptions 1-4 there exists a base learning rate η under which for Algorithm 3 we have*

$$\begin{aligned}&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\bar{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \le \mathcal{O}\left(\sqrt{\frac{(G_{\text{diff}}))\,(1+\epsilon_\eta)}{N\tau T}}\right)\\ &+ \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}N^{1/3}}{p^{2/3}T^{2/3}}(1+\epsilon_\eta^{1/3})\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})^{2/3}N^{1/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right)\\ &+ \mathcal{O}\left(\frac{N\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G_{\text{diff}})}{T}\right) + \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\epsilon_b\right)\end{aligned}$$

Here $G_{\text{diff}} = G(\bar{\mathbf{x}}^{(0,0)}) - G_{\text{inf}}$,is the initial distance from the infimum and, $\mathcal{O}$ subsumes all constants. The parameter τ denotes the number of local steps performed in each iteration. The terms $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ and $\epsilon_b = (1-p)^{\tau_b}K_{\max}(N-1)\sum_{k=1}^K \lambda_k^2$, represent the learning rate estimation error and the pre-factor estimation error, respectively. Here, τ_η is the number of gossip averaging steps used for estimating the scaling factors, and τ_b is the number of steps of log *gossip used to estimate the product terms $M_{ki}^{(t)}$, $K_{\max} = \max_i |\mathcal{K}_i|$, the maximum number of groups a single node is the part of. The initial consensus error is defined as $\Xi_0^2 = \frac{1}{N}\sum_{i=1}^N \|\bar{\mathbf{x}}^{(0,0)} - \mathbf{x}_i^{(0,0)}\|^2$.*

See Appendix D for the proof. The terms that do not vanish as $T \to \infty$ are the learning rate estimation error $\epsilon_\eta = \frac{N^2(1-p)^{\tau_\eta}}{4}$ and pre-factor estimation error $\epsilon_b = (1-p)^{\tau_b}K_{\max}(N-1)\sum_{k=1}^K \lambda_k^2$. These terms, however, can be made arbitrarily small by increasing τ_η and τ_b, independent of the other terms with negligible communication overhead (see Remark 4). We observe a linear speedup in both the number of clients N and the number of local steps τ for the dominant term ($\mathcal{O}(T^{-1/2})$), which decays the slowest with the number of iterations T. There is a slowdown dependent on N in some of the faster decaying non-dominant terms, this happens because in the group wise penalty the gradient to be computed

at one client can potentially depend on the state of other clients. However for large values of T these terms decay fast and have less of an effect on convergence.

B Useful inequalities

PROPOSITION 3. *For a set of n vectors $\{\mathbf{w}_i\}_{i=1}^n$, where $\mathbf{w}_i \in \mathbb{R}^d$, the following inequality holds:*

$$\left\|\sum_{i=1}^n \mathbf{w}_i\right\|^2 \leq n \sum_{i=1}^n \|\mathbf{w}_i\|^2.$$

Moreover, for any two vectors $\mathbf{w}, \mathbf{v} \in \mathbb{R}^d$, we have:

$$\|\mathbf{w}+\mathbf{v}\|^2 \leq (1+\alpha)\|\mathbf{w}\|^2 + (1+\alpha^{-1})\|\mathbf{v}\|^2, \quad \forall \alpha > 0.$$

This inequality also extends to matrices $W, V \in \mathbb{R}^{n\times d}$ in the Frobenius norm.

PROPOSITION 4 (ADAPTED VERSION OF LEMMA 4 IN [15]). *Let $\{X_i\}_{i=1}^N$ be N random variables in $\mathbb{R}^d$, which may not necessarily be independent. Suppose that the conditional expectation satisfies $\mathbb{E}[X_i|X_{i-1},\ldots,X_1] = Y_i$, and that the variance condition $\mathbb{E}[\|X_i - Y_i\|_2^2] \leq \sigma_i^2$ holds. Then, we have*

$$\mathbb{E}\left[\left\|\sum_{i=1}^N (X_i - Y_i)\right\|_2^2\right] \leq \sum_{i=1}^N \sigma_i^2.$$

This result is used while bounding the variance of local gradient updates.

LEMMA 1 (TUNING THE LEARNING RATE - LEMMA 17 IN [16]). *For any non-negative parameters A_0, A_2, A_3, A_4, there exists a constant learning rate η satisfying $\eta \leq \frac{1}{A_4}$ such that*

$$\Psi_T = \frac{A_0}{\eta T} + A_2\eta + A_3\eta^2 \leq 2\left(\frac{A_2 A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4 A_0}{T}.$$

The learning rate is chosen as $\eta = \min\left\{\left(\frac{A_0}{A_2 T}\right)^{\frac{1}{2}}, \left(\frac{A_0}{A_3 T}\right)^{\frac{1}{3}}, \frac{1}{A_4}\right\} \leq \frac{1}{A_4}$.

C Modified Objective with Node-wise Penalty

In this section, we define the following terms, which will be used exclusively within this context.

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \ldots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$

$$\bar{\mathbf{X}}^{(\mathbf{t},\mathbf{0})} := \left[\bar{\mathbf{x}}^{(t,0)}, \ldots, \bar{\mathbf{x}}^{(t,0)}\right] = \mathbf{X}^{(t,0)} \cdot \frac{1}{N}\mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{d\times N}$$

where $\bar{\mathbf{x}}^{(t,0)} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i^{(t,0)}$. The modified objective function with node wise penalty is defined as follows

$$G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N G_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma)F_i(\mathbf{x}) + \gamma a_i \sigma_i(\mathbf{x})) \tag{10}$$

and its gradient at $\mathbf{x}$ is given by $\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N \nabla G_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma) + \gamma a_i \sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x})))\nabla F_i(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N ((1-\gamma) + \gamma a_i q_i(\mathbf{x}))\nabla F_i(\mathbf{x})$ where $q_i(\mathbf{x}) = \sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x}))$. We also define the aggregated update at client i and its noiseless value as

$$\mathbf{h}_i^t = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) \tag{11}$$

$$\bar{\mathbf{h}}_i^t = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}) = b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)}). \tag{12}$$

Here $b_i^{(t)} = (1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i$. Note that $\nabla G_i(\mathbf{x}_i^{(t,0)}) = \left((1-\gamma) + \gamma q_i(\mathbf{x}_i^{(t,0)})a_i\right)\nabla F_i(\mathbf{x}_i^{(t,0)}) = \nabla b_i^{(t)} F_i(\mathbf{x}_i^{(t,0)})$. Define $\tilde{\eta}^{(t)} := \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ which is the effective learning rate w.r.t. the modified objective G at $(t,0)$ and $\tilde{\eta}_i^{(t)} := \frac{\eta}{s_i^{(t)}+\epsilon}$ is the estimate of $\tilde{\eta}^{(t)}$ at node $i \in [N]$. Note that the update at node i during round r can be rewritten as

$$\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = \mathbf{x}_i^{(t,r)} - \tilde{\eta}_i^{(t)} b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})/\tau. \tag{13}$$

For $a_{\max} = \max_i a_i$ we define the constant

$$w_{\max} = 4(1-\gamma) + a_{\max}\gamma. \tag{14}$$

We shall repeatedly used the inequality $(1-\gamma)+\gamma a_i q_i(\mathbf{x}) \leq (1-\gamma)+\gamma a_{\max} q_i(\mathbf{x}) \leq (1-\gamma)+\frac{\gamma a_{\max}}{4} = \frac{w_{\max}}{4}$. We also have the pre-factor $b_i^{(t)} \leq \frac{w_{\max}}{4}\ \forall i, t$. From our update rule and $\tilde{L}_s$ smoothness of function G from lemma 2 we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\underbrace{\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\rangle\right]}_{T_1} + \underbrace{\mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\|^2\right]}_{T_2}. \tag{15}$$

We first focus on simplifying the term T_2.

$$T_2 = \mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\mathbf{h}_i^t\right)\right\|^2\right] \tag{16}$$

$$= \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2}{2}\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\mathbf{h}_i^t\right)\right\|^2\right] \tag{17}$$

$$\leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right)\right\|^2\right] + \underbrace{\tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t\right)\right)\right\|^2\right]}_{T_3} \tag{18}$$

T_3 here can be bounded as

$$T_3 = \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t\right)\right)\right\|^2\right] \tag{19}$$

$$= \frac{\tilde{L}_s}{\tau^2 N^2}\mathbb{E}\left[\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2\left\|\left(b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) - F_i(\mathbf{x}_i^{(t,r)})\right)\right)\right\|^2\right] \tag{20}$$

$$\leq \frac{\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2 \tag{21}$$

$$\leq \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2 + \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}\|\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)}\|^2(b_i^{(t)})^2\sigma^2 \tag{22}$$

$$\leq \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2\epsilon_\eta}{8\tau N}. \tag{23}$$

Here equation (21) follows from proposition 4 and equation 20 follows from independent sampling of mini-batches by the clients, (23) follows from the fact $\tilde{\eta}^{(t)}b_i^{(t)} \leq \eta$, $b_i^{(t)} \leq \frac{w_{\max}}{4}$ and lemma 3. Thus we have

$$T_2 \leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right)\right\|^2\right] + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N}(1+\epsilon_\eta). \tag{24}$$

Consider the term T_1

$$T_1 = -\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\bar{\mathbf{h}}_i^t\right)\right\rangle\right] - \mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}(\mathbf{h}_i^t - \bar{\mathbf{h}}_i^t)\right)\right\rangle\right] \tag{25}$$

$$= -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)})\right\|^2 - \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2}_{T_4} + \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2}_{T_5}. \tag{26}$$

We can bound T_5 as follows

$$T_5 = \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\bar{\mathbf{h}}_i^t\right\|^2 \tag{27}$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\bar{\mathbf{h}}_i^t\right\|^2 \tag{28}$$

$$+ \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\nabla G_i(\bar{\mathbf{x}}^{(t,0)})-\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{29}$$

$$+ \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2$$

$$\leq \frac{3\tilde{\eta}^{(t)}\tilde{L}_s^2}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{30}$$

$$+ \frac{3\tilde{\eta}^{(t)}L_c^2}{2N}\sum_{i=1}^{N}\frac{\|\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2$$

$$\leq \frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2 + \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2 \tag{31}$$

$$+ \frac{3}{2}L_c^2\tilde{\eta}^{(t)}\max_i(b_i^{(t)})^2\epsilon_\eta$$

$$\leq \underbrace{\frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2N}\sum_{i=1}^{N}\mathbb{E}\left\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\right\|^2}_{\text{Consensus Error}} + \underbrace{\frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2}_{T_6} + \frac{3L_c^2\tilde{\eta}^{(t)}w_{\max}^2\epsilon_\eta}{32}. \tag{32}$$

Here equation (31) follows from lemma 4 and equation (32) follows from $\tilde{\eta}^{(t)}b_i^{(t)} \leq \eta$ and $b_i^{(t)} \leq \frac{w_{\max}}{4}$. We can bound T_6 as follows

$$T_6 = \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\left(\nabla G_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau}\overline{\mathbf{h}}_i^t\right)\right\|^2 \tag{33}$$

$$= \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}b_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,0)})-\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 \tag{34}$$

$$= \frac{3\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{N\tau}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,0)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2 \tag{35}$$

$$\leq \frac{3\tilde{\eta}^{(t)}}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2\right] \tag{36}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \tag{37}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\Bigg(4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2\right] + 2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]$$

$$+ 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\Bigg) \tag{38}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}\Bigg(\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2\right]}{4}$$

$$+\frac{w_{\max}^2\tau(L_c^2)\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]}{8} + \frac{3w_{\max}^4(\tilde{\eta}^{(t)})^2\sigma^2}{128} + \frac{3w_{\max}^2(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2}{8}\Bigg) \tag{39}$$

$$\leq \frac{3\tilde{\eta}^{(t)}L_s^2}{2N\tau}\sum_{i=1}^{N}\left(\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2\tau\mathbb{E}\left[\left\|\nabla G_i(\mathbf{x}_i^{(t,0)})\right\|^2\right]}{4}\right) + \frac{3\tilde{\eta}^{(t)}L_s^2}{256N\tau}\left(w_{\max}^4(\tilde{\eta}^{(t)})^2\tau L_c^2N\epsilon_\eta + 3Nw_{\max}^4(\tilde{\eta}^{(t)})^2\sigma^2(1+\epsilon_\eta)\right) \tag{40}$$

$$= \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{8N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta) \tag{41}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right) - \nabla G_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] \tag{42}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\nabla G_i\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right] \tag{43}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\beta}^2}{4} \mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 L_c^2}{256}\epsilon_\eta + \frac{9(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^4 \sigma^2}{256\tau}(1+\epsilon_\eta)$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right] + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\kappa}^2}{4}. \tag{44}$$

Here (39) follows from $b_i^{(t)} \leq \frac{w_{\max}}{4}$ and equation (38) follows from lemma 5. Equation (40) follows from lemma 3 and the definition of $G(\mathbf{x})$. Now from the equations (24), (26), (31) and (44) for $\Xi_t^2 := \frac{1}{N}\sum_{i=1}^{N} E\left[\left\|\overline{\mathbf{x}}^{(t,0)} - \mathbf{x}_i^{(t,0)}\right\|^2\right]$ we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2 + \tilde{\eta}^{(t)}\left(\tilde{L}_s\tilde{\eta}^{(t)} - \frac{1}{2}\right)\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\overline{\mathbf{h}}_i^t\right)\right\|^2\right]$$

$$+ \left(\frac{3\tilde{L}_s^2\tilde{\eta}^{(t)}}{2} + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{L}_s^2}{4}\right)\Xi_t^2 + \left(\frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{9(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2}{256\tau}\right)\sigma^2$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 L_s^2 w_{\max}^2 \tilde{\kappa}^2}{4} + \left(\frac{3L_c^2\tilde{\eta}^{(t)} w_{\max}^2\left(8 + (\tilde{\eta}^{(t)})^2 w_{\max}^2 L_s^2\right)}{256} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau N} + \frac{9(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2\sigma^2}{256\tau}\right)\epsilon_\eta$$

$$+ \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^2 L_s^2 \tilde{\beta}^2}{4}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right]. \tag{45}$$

Using $\tilde{\eta}^{(t)} \leq \min\left\{\frac{1}{4\tilde{L}_s}, \frac{\sqrt{2}}{\sqrt{3}L_s}\right\}$ we can simplify further

$$\tilde{\eta}^{(t)}\left(\frac{1}{2} - \frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2\tilde{\beta}^2}{4}\right)\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] + 2\tilde{L}_s^2\tilde{\eta}^{(t)}\Xi_t^2$$

$$+\tilde{\eta}^{(t)}\frac{3L_c^2 w_{\max}^2}{32}\epsilon_\eta + (\tilde{\eta}^{(t)})^2\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N}\right) + (\tilde{\eta}^{(t)})^3\left(\frac{3L_c^2 L_s^2 w_{\max}^4}{256}\epsilon_\eta + \frac{9L_s^2 w_{\max}^4\sigma^2}{256\tau}(1+\epsilon_\eta) + \frac{3L_s^2 w_{\max}^2\tilde{\kappa}^2}{4}\right). \tag{46}$$

Using the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ and averaging over T we have

$$\frac{1}{T}\sum_{t=1}^{T}\left(\tilde{\eta}^{(t)}\left(\frac{1}{2} - \frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2\tilde{\beta}^2}{4}\right)\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right]\right) \leq \frac{\sum_{t=1}^{T}\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right]}{T} + \frac{2\tilde{L}_s^2\eta}{\epsilon T}\sum_{t=1}^{T}\Xi_t^2$$

$$+\frac{3\eta L_c^2 w_{\max}^2}{32\epsilon}\epsilon_\eta + (\eta)^2\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\epsilon^2}\right) + (\eta)^3\left(\frac{3L_c^2 L_s^2 w_{\max}^4}{256\epsilon^3}\epsilon_\eta + \frac{9L_s^2 w_{\max}^4}{256\epsilon^3\tau\sigma^2}(1+\epsilon_\eta) + \frac{3L_s^2 w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\right). \tag{47}$$

Now using lemma 6 we have

$$\frac{1}{T}\sum_{t=1}^{T}\tilde{\eta}^{(t)}\left[\frac{1}{2} - \tilde{\beta}^2\left(\frac{3L_s^2(\tilde{\eta}^{(t)})^2 w_{\max}^2}{4} + \frac{256\tilde{L}_s^2\eta}{p^2\epsilon}\tilde{\eta}^{(t)}\right)\right]\mathbb{E}\left[\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2\right] \leq \frac{G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}}{T}$$

$$+\frac{2\tilde{L}_s^2\eta}{\epsilon}\left\{\frac{4\Xi_0^2}{pT}+\frac{128}{p^2T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2+\frac{2w_{\max}^2}{p^2T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n+\frac{3w_{\max}^2}{p^2\tau T}\sum_{t=1}^{T}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta)\right\}$$

$$+\frac{3\eta L_c^2 w_{\max}^2}{32\epsilon}\epsilon_\eta+(\eta)^2\Big(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\epsilon^2}\Big)+(\eta)^3\Big(\frac{3L_c^2L_s^2w_{\max}^4}{256\epsilon^3}\epsilon_\eta+\frac{9L_s^2w_{\max}^4\sigma^2}{256\epsilon^3\tau}(1+\epsilon_\eta)+\frac{3L_s^2w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\Big). \tag{48}$$

Again using the fact $\frac{\eta}{\epsilon+\frac{Nw_{\max}}{4}}\le\tilde{\eta}^{(t)}\le\frac{\eta}{\epsilon}$ for $\eta\le\frac{\epsilon}{2\tilde{\beta}\sqrt{\left(\frac{L_s^2w_{\max}^2}{4}+\frac{256\tilde{L}_s^2}{p^2}\right)}}$ we have $\frac{1}{2}-\tilde{\beta}^2\left(\frac{L_s^2(\tilde{\eta}^{(t)})^2w_{\max}^2}{4}+\frac{256\tilde{L}_s^2\eta}{p^2\epsilon}\tilde{\eta}^{(t)}\right)\le\frac{1}{4}$ and the LHS can be lower bounded by $\frac{\eta}{4(\epsilon+Nw_{\max}/4)}\left(\frac{1}{T}\sum_{t=0}^{T-1}\left(\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\right)\right)$. Further substituting $\tilde{\eta}^{(t)}\le\frac{\eta}{\epsilon}$ in the RHS we have

$$\frac{\eta}{4(\epsilon+Nw_{\max}/4)}\left(\frac{1}{T}\sum_{t=0}^{T-1}\left(\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\right)\right)\le\frac{G(\bar{\mathbf{x}}^{(0,0)})-G_{\inf}}{T}$$

$$+\left(\frac{8\tilde{L}_s^2\,\Xi_0^2}{pT\,\epsilon}+\frac{3L_c^2w_{\max}^2}{32\epsilon}\epsilon_\eta\right)\eta+\left(\frac{\tilde{L}_s w_{\max}^2\sigma^2(1+\epsilon_\eta)}{8\tau N\,\epsilon^2}\right)\eta^2+$$

$$\left(\frac{256\tilde{L}_s^2\tilde{\kappa}^2}{p^2\,\epsilon^3}+\frac{4\tilde{L}_s^2w_{\max}^2L_c^2\,\epsilon_n}{p^2\,\epsilon^3}+\frac{6\tilde{L}_s^2w_{\max}^2\sigma^2}{p^2\tau\,\epsilon^3}(1+\epsilon_\eta)+\frac{3L_c^2L_s^2w_{\max}^4}{256\epsilon^3}\epsilon_\eta+\frac{9L_s^2w_{\max}^4\sigma^2}{256\epsilon^3\tau}(1+\epsilon_\eta)+\frac{3L_s^2w_{\max}^2\tilde{\kappa}^2}{4\epsilon^3}\right)\eta^3. \tag{49}$$

Choosing $\epsilon=\frac{N\gamma a_{\max}L_c^2}{6\sqrt{3}L_s}$ (see lemma 2) for making the step size $\tilde{\eta}^{(t)}$ inversely proportional to the smoothness parameter of the global objective G we can rewrite equation 49 as

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\le\frac{A_0}{\eta T}+A_1+A_2\eta+A_3\eta^2 \tag{50}$$

where

$$A_0=\frac{N\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{3\sqrt{3}\,L_s}\Big[G(\bar{\mathbf{x}}^{(0,0)})-G_{\inf}\Big], \tag{51}$$

$$A_1=\frac{16\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,\tilde{L}_s^2\,\Xi_0^2}{pT\,\gamma\,a_{\max}L_c^2}+\frac{3\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,w_{\max}^2}{16\,\gamma\,a_{\max}}\,\epsilon_\eta, \tag{52}$$

$$A_2=\frac{9\,\tilde{L}_s\,w_{\max}^2\,\sigma^2\,(1+\epsilon_\eta)\,L_s\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{2\sqrt{3}\,\tau\,N^2\,\gamma^2a_{\max}^2L_c^4}, \tag{53}$$

$$A_3=\frac{216\,L_s^2\,(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)}{N^2\,\gamma^3a_{\max}^3L_c^6}\left[\frac{256\,\tilde{L}_s^2\,\tilde{\kappa}^2}{p^2}+\frac{4\,\tilde{L}_s^2\,w_{\max}^2\,L_c^2\,\epsilon_n}{p^2}+\frac{6\,\tilde{L}_s^2\,w_{\max}^2\,\sigma^2}{p^2\tau}(1+\epsilon_\eta)\right.$$

$$\left.+\frac{3\,L_c^2\,L_s^2\,w_{\max}^4}{256}\,\epsilon_\eta+\frac{9\,L_s^2\,w_{\max}^4\,\sigma^2}{256\tau}(1+\epsilon_\eta)+\frac{3\,L_s^2\,w_{\max}^2\,\tilde{\kappa}^2}{4}\right]. \tag{54}$$

For small values of gamma we can use the fact $\tilde{\eta}^{(t)}\le\frac{\eta}{N(1-\gamma)}$ and using the same steps as above starting from equation (45) we obtain the bound

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\overline{\mathbf{x}}^{(t,0)}\big)\right\|^2\right]\le\frac{A_0}{\eta T}+A_1+A_2\eta+A_3\eta^2 \tag{55}$$

for

$$A_0=\frac{N\Big(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s\Big)}{3\sqrt{3}\,L_s}\Big[G\big(\bar{\mathbf{x}}^{(0,0)}\big)-G_{\inf}\Big], \tag{56}$$

$$A_1=\frac{2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s}{3\sqrt{3}\,L_s}\left[\frac{8\,\tilde{L}_s^2\,\Xi_0^2}{pT\,(1-\gamma)}+\frac{3L_c^2\,w_{\max}^2}{32\,(1-\gamma)}\,\epsilon_\eta\right], \tag{57}$$

$$A_2=\frac{(2\gamma\,a_{\max}L_c^2+3\sqrt{3}\,w_{\max}L_s)\,\tilde{L}_s\,w_{\max}^2\,\sigma^2\,(1+\epsilon_\eta)}{24\sqrt{3}\,L_s\,\tau\,N^2\,(1-\gamma)^2}, \tag{58}$$

$$A_3 = \frac{2\gamma\, a_{\max} L_c^2 + 3\sqrt{3}\, w_{\max} L_s}{3\sqrt{3}\, L_s N^2\,(1-\gamma)^3}\Bigg[\frac{256\,\tilde{L}_s^2\,\tilde{\kappa}^2}{p^2} + \frac{4\,\tilde{L}_s^2\, w_{\max}^2\, L_c^2\,\epsilon_n}{p^2} + \frac{6\,\tilde{L}_s^2\, w_{\max}^2\,\sigma^2}{p^2\tau}(1+\epsilon_\eta) + \frac{3L_c^2\, L_s^2\, w_{\max}^4}{256}\,\epsilon_\eta + \frac{9L_s^2\, w_{\max}^4\,\sigma^2}{256\tau}(1+\epsilon_\eta) + \frac{3L_s^2\, w_{\max}^2\,\tilde{\kappa}^2}{4}\Bigg]. \tag{59}$$

The final bound is obtained using lemma 1, taking the case wise minimum of the two expressions and choosing $A_4 = \max\left\{\frac{2\tilde{\beta}\sqrt{\left(\frac{L_s^2 w_{\max}^2}{4} + \frac{256\tilde{L}_s^2}{p^2}\right)}}{\epsilon}, \frac{4\tilde{L}_s}{\epsilon}, \frac{3}{\sqrt{2}\epsilon}\right\}$, we have for $\epsilon = \frac{N\gamma a_{\max} L_c^2}{6\sqrt{3}L_s}$ chosen as per lemma (2) to make the step size $\tilde{\eta}^{(t)}$ with respect to the modified objective G inversely proportional to the smoothness parameter

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\big(\bar{\mathbf{x}}^{(t,0)}\big)\right\|^2\right] \le A_1 + 2\left(\frac{A_2 A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4 A_0}{T} \tag{60}$$

$$\begin{aligned}
&= \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\sqrt{\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)\,(1+\epsilon_\eta)}{N\tau T}}\right) \\
&+ \mathcal{O}\left(\frac{\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf})}{T}\right) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{T^{2/3}}\epsilon_\eta^{1/3}\Big) + \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\Big) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}T^{2/3}}\Big) + \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}T^{2/3}}\epsilon_\eta^{1/3}\Big) \\
&+ \mathcal{O}\Big(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\Big) + \mathcal{O}\left(\frac{\big(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\big)^{2/3}}{T^{2/3}}\right).
\end{aligned} \tag{61}$$

C.1 Auxiliary Lemmas

LEMMA 2 (SMOOTHNESS OF THE MODIFIED OBJECTIVE). *Under Assumption 2 the modified local objectives $G_1(\mathbf{x})\cdots G_N(\mathbf{x})$ are Lipschitz smooth for any $\mathbf{x}$ with the smoothness parameter given by $\frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + ((1-\gamma) + \gamma a_{\max} q_i(\mathbf{x}))L_s \le \frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + ((1-\gamma) + \frac{\gamma a_{\max}}{4})L_s = \frac{\gamma a_{\max} L_c^2}{6\sqrt{3}} + \frac{w_{\max}}{4}L_s = \tilde{L}_s$.*

PROOF. Let $\|.\|_{op}$ denote the spectral norm of the matrix. We need to show a finite upper bound on the operator norm of Hessian of G for $\tilde{L}_s$ smoothness. At any point $\mathbf{x}$, for $G_i(\mathbf{x}) = ((1-\gamma)F_i(\mathbf{x}) + \gamma a_i \sigma_i(\mathbf{x}))$ we have

$$\begin{aligned}
&\|\nabla^2 G_i(\mathbf{x})\|_{op} = \|(1-\gamma)\nabla^2 F_i(\mathbf{x}) + \gamma a_i \nabla^2\big(\sigma_i(\mathbf{x})\big)\|_{op} \\
&= \|\gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x}))(1-2\sigma(\mathbf{x}))(\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^\top) + ((1-\gamma) + \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x})))\nabla^2 F_i(\mathbf{x})\|_{op} \\
&\le \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x}))(1-2\sigma(\mathbf{x}))\left\|\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^\top\right\|_{op} + ((1-\gamma) + \gamma a_i \sigma(\mathbf{x})(1-\sigma(\mathbf{x})))\|\nabla^2 F_i(\mathbf{x})\|_{op} \\
&\le \frac{\gamma a_i L_c^2}{6\sqrt{3}} + ((1-\gamma) + \gamma a_i q_i(\mathbf{x}))L_s.
\end{aligned}$$

Replacing a_i by $a_{\max}$ gives us a uniform bound for all $i \in [N]$. Further bounding $\gamma a_{\max}$ by $w_{\max}$ we can bound the L-smoothness parameter for the global objective G by

$$\frac{w_{\max} L_c^2}{6\sqrt{3}} + \frac{L_s}{N}\sum_{i=1}^{N}((1-\gamma) + \gamma a_i q_i(\mathbf{x})). \tag{62}$$

□

PROPOSITION 5 (BOUNDED DISSIMILARITY OF $G(\mathbf{x})$). *Under Assumptions 2 and 3 there exist constants $\tilde{\beta}$ and $\tilde{\kappa}$ such that*

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla G_i(\mathbf{x})\|^2 \le \tilde{\beta}^2\|\nabla G(\mathbf{x})\|^2 + \tilde{\kappa}^2. \tag{63}$$

PROOF. One can show that

$$\frac{1}{N}\sum_{i=1}^{N}\|\nabla G_i(\mathbf{x})\|^2 = \frac{1}{N}\sum_{i=1}^{N}((1-\gamma)+\gamma a_i q_i(\mathbf{x}))^2\,\|\nabla F_i(\mathbf{x})\|^2 \tag{64}$$

$$\leq \frac{w_{\max}^2}{16N}\sum_{i=1}^{N}\|\nabla F_i(\mathbf{x})\|^2 \tag{65}$$

$$\leq \beta^2\left\|\frac{w_{\max}}{4}\nabla F(\mathbf{x})\right\|^2 + \frac{w_{\max}^2\kappa^2}{16} \tag{66}$$

$$\leq 2\beta^2\|\nabla G(\mathbf{x})\|^2 + 2\beta^2\left\|\nabla G(\mathbf{x}) - \frac{w_{\max}}{4}\nabla F(\mathbf{x})\right\|^2 + \frac{w_{\max}^2\kappa^2}{16} \tag{67}$$

$$\leq 2\beta^2\|\nabla G(\mathbf{x})\|^2 + \frac{\beta^2 w_{\max}^2 L_c^2}{4} + \frac{w_{\max}^2\kappa^2}{16}. \tag{68}$$

Thus the inequality will always hold for $\tilde{\beta} = 2\beta^2$ and $\tilde{\kappa}^2 = \frac{\beta^2 w_{\max}^2 L_c^2}{4} + \frac{w_{\max}^2\kappa^2}{16}$. □

LEMMA 3 (LEARNING RATE ESTIMATION ERROR). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (b_i^{(t)})^2 \leq \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{69}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2\epsilon_\eta}{16}\right\} \tag{70}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta}N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. We have $\tilde{\eta}_i^{(t)} = \frac{\eta}{s_i^{(t)}+\epsilon}$ and $\tilde{\eta}^{(t)} = \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ where $s_i^{(t)} = \max\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\}$, $s_i^{(t,0)} = Nb_i^{(t)}$, and $s_i^{(t,r+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\, s_j^{(t,r)}$, or equivalently $\mathbf{s}^{(t,r+1)} = \mathbf{W}\mathbf{s}^{(t,r)}$, with $\mathbf{s}^{(t,r)} = [s_1^{(t,r)}\cdots s_N^{(t,r)}]$. Also, $\bar{s}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} s_i^{(t,r)}$ is the same for all r since gossip averaging preserves the sum, thus $\bar{s}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} s_i^{(t,0)} = \sum_{i=1}^{N} b_i^{(t)}$. We bound the estimation error as follows.

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (b_i^{(t)})^2 = \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\left\|\frac{\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}}{\tilde{\eta}^{(t)}}\right\|^2 \tag{71}$$

$$= \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\frac{\left\|\frac{1}{s_i^{(t)}+\epsilon} - \frac{1}{\bar{s}^{(t)}+\epsilon}\right\|^2}{\left\|\frac{1}{\bar{s}^{(t)}+\epsilon}\right\|^2} \tag{72}$$

$$= \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}(b_i^{(t)})^2\frac{\left\|s_i^{(t)} - \bar{s}^{(t)}\right\|^2}{\left\|s_i^{(t)} + \epsilon\right\|^2} \tag{73}$$

$$\leq \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t)} - \bar{s}^{(t)}\right\|^2 \leq \frac{\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\right\|^2 \tag{74}$$

$$\leq \frac{(1-p)^{\tau_\eta}\left\|\tilde{\eta}^{(t)}\right\|^2}{N}\sum_{i=1}^{N}\left\|s_i^{(t,0)} - \bar{s}^{(t)}\right\|^2 \tag{75}$$

$$\leq \frac{(1-p)^{\tau_\eta}\left\|\tilde{\eta}^{(t)}\right\|^2(\max_i s_i^{(t,0)})^2}{4} \tag{76}$$

$$\leq \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{77}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2\epsilon_\eta}{16}\right\} \tag{78}$$

Here equation 74 follows since $s_i^{(t)} \geq b_i^{(t)} = s_i^{(t,0)}/N$ and for the second part note if $s_i^{(t)} = s_i^{(t,\tau_\eta)}$ we have $\|s_i^{(t)} - \bar{s}^{(t)}\|^2 = \|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\|^2$ and if $s_i^{(t)} = s_i^{(t,0)}/N$, we have $s_i^{(t,\tau_\eta)} \leq s_i^{(t,0)}/N \leq \bar{s}^{(t)}$. since $\bar{s}^{(t)} \geq b_i^{(t)} = s_i^{(t,0)}/N$, hence $\|s_i^{(t)} - \bar{s}^{(t)}\|^2 \leq \|s_i^{(t,\tau_\eta)} - \bar{s}^{(t)}\|^2$ in both cases. Equation (76) follows from $\sum_{i=1}^N \|c_i - \bar{c}\|^2 \leq \frac{N \max_i (c_i)^2}{4}$ for $\bar{c} = \frac{1}{N}\sum_{i=1}^N c_i,\ c_i \geq 0$ and equation (77) follows from $\tilde{\eta}^{(t)} s_i^{(t,0)} = N\left(\frac{\eta}{s_i^{(t)}+\epsilon}\right) b_i^{(t)} \leq N\eta\ \forall i$. □

LEMMA 4 (FRACTIONAL LEARNING RATE ESTIMATION ERROR). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^N \frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2 \leq \frac{(1-p)^{\tau_\eta} N^2 \max_i (b_i^{(t)})^2}{4} = \max_i (b_i^{(t)})^2 \epsilon_\eta \tag{79}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta} N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. Proceeding similarly as lemma 3 until equation (76) we have

$$\frac{1}{N}\sum_{i=1}^N \frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(b_i^{(t)})^2 \leq \frac{(1-p)^{\tau_\eta} (\max_i s_i^{(t,0)})^2}{4} = \frac{(1-p)^{\tau_\eta} N^2 \max_i (b_i^{(t)})^2}{4} = \max_i (b_i^{(t)})^2 \epsilon_\eta. \tag{80}$$

□

LEMMA 5 (BOUNDING DRIFT). *For client i, the accumulated local drift error over τ local steps in Algorithm 2 can be bounded as:*

$$\begin{aligned}\sum_{r=1}^{\tau} \mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq 4(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \tau \mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + 2\tau L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] \\ &\quad + 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 \sigma^2\right].\end{aligned} \tag{81}$$

PROOF.

$$\tau^2 \mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right] = \mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right)\right\|^2\right] \tag{82}$$

$$\begin{aligned}&\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \\ &\quad + 3\mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right) - \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right)\right\|^2\right]\end{aligned} \tag{83}$$

$$\begin{aligned}&\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3r^2 L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] \\ &\quad + 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 \sum_{l=0}^{r-1} \underbrace{\left\|\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}, \xi_i^{(t,l)}\right) - \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right)\right\|^2}_{\leq \sigma^2}\right]\end{aligned} \tag{84}$$

$$\leq \underbrace{3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\, b_i^{(t)} \sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]}_{T_7} + 3r^2 L_c^2 \mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right) b_i^{(t)}\right\|^2\right] + 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2 (b_i^{(t)})^2 r\sigma^2\right] \tag{85}$$

Here equation (85) follows from proposition (4). We can bound T_7 as

$$T_7 = 3(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \mathbb{E}\left[\left\|\sum_{l=0}^{r-1} \nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \tag{86}$$

$$\leq r \sum_{l=0}^{r-1} 3(\tilde{\eta}^{(t)})^2 (b_i^{(t)})^2 \mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] \tag{87}$$

$$\leq r\sum_{l=0}^{r-1} 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\right) \tag{88}$$

$$\leq r\sum_{l=0}^{r-1} 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,l)}-\mathbf{x}_i^{(t,0)}\right\|^2\right]\right) \tag{89}$$

$$\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+r\sum_{l=0}^{r-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,l)}-\mathbf{x}_i^{(t,0)}\right\|^2\right]\right) \tag{90}$$

$$\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau\sum_{l=0}^{\tau-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,l)}\right\|^2\right]\right). \tag{91}$$

Substituting equation (91) into equation (85) we have

$$\begin{aligned}\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq 6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\left(r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau\sum_{l=0}^{\tau-1}L_s^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,l)}\right\|^2\right]\right)\\ &\quad+3r^2L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2r\sigma^2\right].\end{aligned} \tag{92}$$

Summing over τ

$$\begin{aligned}\tau^2\left(1-6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2L_s^2\right)\sum_{i=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] &\leq \sum_{r=1}^{\tau}6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2r^2\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\\ &\quad+\sum_{r=1}^{\tau}3r^2L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+\sum_{r=1}^{\tau}3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2r\sigma^2\right]\end{aligned} \tag{93}$$

$$\leq 2(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau^3\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+\tau^3L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+\frac{3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\tau^2\sigma^2\right]}{2}. \tag{94}$$

Since $\tilde{\eta}^{(t)}(b_i^{(t)})^2 \leq \eta$ for $\eta \leq \frac{1}{2\sqrt{3}L_s}$ we have $\frac{1}{(1-6(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2L_s^2)} \leq \frac{1}{1-6(\eta)^2L_s^2} \leq 2$ and

$$\begin{aligned}&\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq 4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]\\ &+2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]+3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\end{aligned} \tag{95}$$

$$\begin{aligned}&\leq 4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right]+2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]\\ &+6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right]+6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right].\end{aligned} \tag{96}$$

□

LEMMA 6 (CONSENSUS RECURSION). *Define* $\Xi_t^2 := \frac{1}{N}\sum_{i=1}^{N}\|\bar{\mathbf{x}}^{(t,0)}-\mathbf{x}_i^{(t,0)}\|^2$. *For* $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ *we have*

$$\Xi_{t+1}^2 \leq \left(1-\frac{p}{4}\right)\Xi_t^2+\frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2+\frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2+\frac{(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{2p}\epsilon_\eta+\frac{3(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{97}$$

PROOF. Let $\mathbf{H}^{(t)} := \frac{1}{\tau}\left[\mathbf{h}_1^t\cdots\mathbf{h}_N^t\right] \in \mathbb{R}^{d\times N}$ and $\tilde{\boldsymbol{\eta}}^{(t)} := [\tilde{\eta}_1^{(t)}\cdots\tilde{\eta}_N^{(t)}] \in \mathbb{R}^{1\times N}$. We have

$$N\Xi_{t+1}^2 = \mathbb{E}\left\|\tilde{\mathbf{X}}^{(t+1,0)}-\mathbf{X}^{(t+1,0)}\right\|_F^2 = \mathbb{E}\left\|\left(\mathbf{X}^{(t+1,0)}\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{98}$$

$$= \mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)W\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{99}$$

$$= \mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\left(W-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{100}$$

$$\leq (1-p)\mathbb{E}\left\|\left(\mathbf{X}^{(t,0)}-\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{101}$$

$$\leq (1-p)\left(1+\frac{p}{2}\right)\mathbb{E}\left\|\mathbf{X}^{(t,0)}\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 + (1-p)\left(1+\frac{2}{p}\right)\mathbb{E}\left\|\left(\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_F^2 \tag{102}$$

$$\leq \left(1-\frac{p}{2}\right)N\Xi_t^2 + \frac{2}{p}\mathbb{E}\left\|\left(\mathbf{H}(\tilde{\boldsymbol{\eta}}^{(t)})^T\right)\right\|_F^2 \tag{103}$$

$$= \left(1-\frac{p}{2}\right)N\Xi_t^2 + \frac{2}{p}\underbrace{\frac{1}{\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2}_{T_8}. \tag{104}$$

Here equation (103) follows from $\|AB\|_F \leq \|A\|_F\|B\|_2$ and $\left\|\left(\mathbf{I}-\frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\right\|_2 \leq 1$. We bound T_8 as

$$T_8 = \frac{1}{\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2 \tag{105}$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)})b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 \tag{106}$$

$$+ 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3}{\tau}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\right\|^2\sigma^2 \tag{107}$$

$$\leq 3\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{6}{\tau}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)}b_i^{(t)}\right\|^2\sigma^2 + \frac{6}{\tau}\sum_{i=1}^{N}\left\|(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)})b_i^{(t)}\right\|^2\sigma^2 \tag{108}$$

$$\leq 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,0)})\right\|^2 + 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$\frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{109}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + 6\sum_{i=1}^{N}\frac{1}{\tau^2}\mathbb{E}\left\|\tilde{\eta}^{(t)}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{110}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{6}{\tau}\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}(\eta)^2\mathbb{E}\left\|\left(\nabla F_i(\mathbf{x}_i^{(t,r)})-\nabla F_i(\mathbf{x}_i^{(t,0)})\right)\right\|^2$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{111}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{6}{\tau}\sum_{i=1}^{N}L_s^2(\eta)^2\sum_{r=0}^{\tau-1}\mathbb{E}\left\|\left(\mathbf{x}_i^{(t,r)}-\mathbf{x}_i^{(t,0)}\right)\right\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta$$

$$+ \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{112}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \frac{6}{\tau}\sum_{i=1}^{N}L_s^2(\eta)^2\Bigg((4(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\tau\mathbb{E}\left[\left\|\nabla F_i\left(\mathbf{x}_i^{(t,0)}\right)\right\|^2\right] + 2\tau L_c^2\mathbb{E}\left[\left\|\left(\tilde{\eta}_i^{(t)}-\tilde{\eta}^{(t)}\right)b_i^{(t)}\right\|^2\right]$$

$$+ 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(b_i^{(t)})^2\sigma^2\right] + 6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(b_i^{(t)})^2\sigma^2\right]\Bigg) \tag{113}$$

$$\leq \sum_{i=1}^{N}6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 24L_s^2(\eta)^2(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\|\nabla G(\mathbf{x}_i^{(t,0)})\|^2\right] + \frac{3NL_s^2(\eta)^2(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{9N(\eta)^2(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_s^2\sigma^2}{4\tau}(1+\epsilon_\eta). \tag{114}$$

Here equation (107) follows from lemma 3 and proposition (4). Equation (109) also uses lemma 3. Equation (110) follows from the definition of G, equation (111) follows from the fact that $\tilde{\eta}^{(t)} b_i^{(t)} \leq \eta$ and equation (113) follows from lemma 5. Equation (114) also follows from from lemma 3. For $\eta \leq \frac{1}{2\sqrt{3}L_s}$ we can upper bound equation (114) as

$$T_8 \leq \sum_{i=1}^{N} 6(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 2(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{16}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16\tau}(1+\epsilon_\eta) \tag{115}$$

$$= \sum_{i=1}^{N} 8(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) \tag{116}$$

$$\leq \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)}) - \nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 \tag{117}$$

$$\leq \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\|\nabla G_i(\mathbf{x}_i^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ \sum_{i=1}^{N} 16(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\|\mathbf{x}_i^{(t,0)} - \overline{\mathbf{x}}^{(t,0)})\|^2 \tag{118}$$

$$= 16\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) + 16N(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2 \tag{119}$$

$$\leq 16N\tilde{\beta}^2(\tilde{\eta}^{(t)})^2\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2 + 16N(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{N(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3N(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta)$$

$$+ 16N(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2. \tag{120}$$

Substituting back in equation (104) we have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{2}\right)\Xi_t^2 + \frac{2}{p}\Bigg(16(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\|^2 + 16(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 \tag{121}$$

$$+ \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{4}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{8\tau}(1+\epsilon_\eta) + 16(\tilde{\eta}^{(t)})^2\tilde{L}_s^2\Xi_t^2\Bigg). \tag{122}$$

For $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ we have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{4}\right)\Xi_t^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{2p}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{123}$$

□

LEMMA 7 (CONSENSUS CONTROL). *For $\tilde{\eta}^{(t)} \leq \frac{\sqrt{p}}{8\sqrt{2}\tilde{L}_s}$ the term $\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2$ can be bounded as*

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \leq \frac{4\Xi_0^2}{pT} + \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\beta}^2\,\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2$$

$$+ \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2 + \frac{2w_{\max}^2}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n + \frac{3w_{\max}^2}{p^2\tau T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta). \tag{124}$$

PROOF. Consider the recursion equation in lemma 6. We have

$$\Xi_{t+1}^2 \leq \left(1 - \frac{p}{4}\right)\Xi_t^2 + A_t \tag{125}$$

where

$$A_t = \left(1 - \frac{p}{4}\right)\Xi_t^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\beta}^2\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 + \frac{32}{p}(\tilde{\eta}^{(t)})^2\tilde{\kappa}^2 + \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 L_c^2}{2p}\epsilon_\eta + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta). \tag{126}$$

Rearranging and summing over T we have

$$\Xi_T^2 - \Xi_0^2 = \sum_{t=0}^{T-1}\left(\Xi_{t+1}^2 - \Xi_t^2\right) \le -\frac{p}{4}\sum_{t=0}^{T-1}\Xi_t^2 + \sum_{t=0}^{T-1}A_t. \tag{127}$$

Since $\Xi_T^2 \ge 0$

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le \frac{4}{pT}\left(\Xi_0^2 + \sum_{t=0}^{T-1}A_t\right). \tag{128}$$

Substituting back A_t gives

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \le{}& \frac{4\Xi_0^2}{pT} + \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\beta}^2\,\|\nabla G_i(\overline{\mathbf{x}}^{(t,0)})\|^2 \\ &+ \frac{128}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\tilde{\kappa}^2 + \frac{2w_{\max}^2}{p^2T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,L_c^2\,\epsilon_n + \frac{3w_{\max}^2}{p^2\tau T}\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2\,\sigma^2(1+\epsilon_\eta).\end{aligned} \tag{129}$$

□

D Modified Objective with Group-wise Penalty

In this section, we define the following terms, which will be used exclusively within this context.

$$\mathbf{X} := [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{d\times N}$$
$$\bar{\mathbf{X}} := [\bar{\mathbf{x}}, \ldots, \bar{\mathbf{x}}] \in \mathbb{R}^{d\times N},$$

where $\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i$. Similarly at the beginning of round t one can define

$$\mathbf{X}^{(t,0)} := \left[\mathbf{x}_1^{(t,0)}, \ldots, \mathbf{x}_N^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$
$$\bar{\mathbf{X}}^{(\mathbf{t},\mathbf{0})} := \left[\bar{\mathbf{x}}^{(t,0)}, \ldots, \bar{\mathbf{x}}^{(t,0)}\right] \in \mathbb{R}^{d\times N},$$

where $\bar{\mathbf{x}}^{(t,0)} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i^{(t,0)}$. The modified objective function with group wise penalty is defined as follows

$$G(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^K \lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)\right)\right\} \tag{130}$$

$$= G(\mathbf{x}) = \frac{1}{N}\left\{(1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\left(\sum_{k=1}^K \lambda_k M_k(\mathbf{x})\right)\right\}. \tag{131}$$

Here $M_k(\mathbf{x}) := \prod_{j\in C_k}\sigma_j(\mathbf{x})$ which tells us about the probability of all nodes in C_k dropping out given a local model $\mathbf{x}$ at each node. . The gradient of the function $\sigma(\mathbf{x})$ is given by $\sigma_i(\mathbf{x})(1-\sigma_i(\mathbf{x}))\nabla F_i(\mathbf{x})$. Note that the gradient of the global objective here is given by

$$\nabla G(\mathbf{x}) = \frac{1}{N}\left(\sum_{i=1}^N(1-\gamma)\nabla F_i(\mathbf{x}) + \gamma\sum_{k=1}^K\sum_{l\in C_k}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_l(\mathbf{x}))\nabla F_l(\mathbf{x})\right). \tag{132}$$

One can rewrite the gradient of the modified objective in equation (9) as

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k M_k(\mathbf{x})(1-\sigma_i(\mathbf{x}))\right)\nabla F_i(\mathbf{x}). \tag{133}$$

Where $\mathcal{K}_i := \{k | i \in C_k\} \subseteq [K]$ is the set of all groups which node i is a part of. For the ease of purpose we define the function $\Psi(\mathbf{X})$ as follows

$$\Psi(\mathbf{X}) := \frac{1}{N}\left((1-\gamma)\left(\sum_{i=1}^N F_i(\mathbf{x})\right) + \gamma\sum_{k=1}^K\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)\right) \tag{134}$$

and its gradient is given by

$$[\nabla\Psi(\mathbf{X})]_i = \frac{1}{N}\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i) \tag{135}$$

$$= \frac{1}{N}\left((1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k(\mathbf{X})(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i) \tag{136}$$

where

$$M_k(\mathbf{X}) = \prod_{j\in C_k} \sigma_j(\mathbf{x}_j)$$

. Let $M_{ki}(\mathbf{X})$ be the estimate of $M_k(\mathbf{X})$ at node i given by

$$M_{ki}(\mathbf{X}) = \min\left\{M_{ki}^{(\tau_b)}(\mathbf{X}), \left(M_{ki}^{(0)}(\mathbf{X})\right)^{1/N}\right\} \tag{137}$$

with

$$M_{ki}^{(0)} = \begin{cases} (\sigma_i(\mathbf{x}_i))^N & \text{if } k \in \mathcal{K}_i \\ 1 & \text{if } k \notin \mathcal{K}_i \end{cases} \tag{138}$$

Here $M_{ki}^{(\tau_b)}(\mathbf{X})$ is the value after τ_b rounds of gossip averaging. Additionally $G(\bar{\mathbf{x}}) = \Psi(\bar{\mathbf{X}})$ and $\nabla G(\bar{\mathbf{x}}) = \sum_{i=1}^N \left[\nabla\Psi(\bar{\mathbf{X}})\right]_i$. We also have $M_{ki}^{(t)} = M_{ki}(\mathbf{X}^{(t,0)})$ and $M_k^{(t)} = M_k(\mathbf{X}^{(t,0)})$. Also recall the terms $b_i^{(t)}$ and $\hat{b}_i^{(t)}$

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k^{(t)}(1-\sigma(\mathbf{x}_i^{(t,0)})) = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_k(\mathbf{X}^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)})) \tag{139}$$

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}^{(t)}(1-\sigma(\mathbf{x}_i^{(t,0)})) = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}(\mathbf{X}^{(t,0)})(1-\sigma(\mathbf{x}_i^{(t,0)})) \tag{140}$$

where $b_i^{(t)}$ denotes the true pre-factor at node i, and $\hat{b}_i^{(t)}$ represents its estimate at node i obtained after τ_b rounds of log gossip. Next we define the aggregated update, its noiseless/expected value and its expected value under perfect pre-factor estimation as

$$\mathbf{h}_i^t = \hat{b}_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) \tag{141}$$

$$\widehat{\mathbf{h}}_i^t = \hat{b}_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}) \tag{142}$$

$$\overline{\mathbf{h}}_i^t = b_i^{(t)} \sum_{r=0}^{\tau-1} \nabla F_i(\mathbf{x}_i^{(t,r)}). \tag{143}$$

We also define the constant

$$w_{\max} = 4(1-\gamma) + \gamma \max_{i\in[N]}\Big(\sum_{k\in\mathcal{K}_i} \lambda_k\Big). \tag{144}$$

We will repeatedly use the fact $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ and $b_i^{(t)} \le \frac{w_{\max}}{4}$ (see lemma 10). Define $\tilde{\eta}^{(t)} := \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ and $\tilde{\eta}_i^{(t)} := \frac{\eta}{s_i^{(t)}+\epsilon}$ is the estimate of $\tilde{\eta}^{(t)}$ at node $i \in [N]$. Note that the update at node i during round r can be rewritten as

$$\mathbf{x}_i^{(t,r+1)} := \mathbf{x}_i^{(t,r)} - \hat{\eta}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)}) = \mathbf{x}_i^{(t,r)} - \tilde{\eta}_i^{(t)} \hat{b}_i^{(t)} \nabla F_i(\mathbf{x}_i^{(t,r)}, \xi_i^{(t,r)})/\tau. \tag{145}$$

From the $\tilde{L}_s$ smoothness of the global objective function (see lemma 8)

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \le \underbrace{-\mathbb{E}\left[\left\langle \nabla G(\bar{\mathbf{x}}^{(t,0)}), \left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\rangle\right]}_{T_9} + \underbrace{\mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\|^2\right]}_{T_{10}}. \tag{146}$$

We first focus on simplifying T_{10}

$$T_{10} = \mathbb{E}\left[\frac{\tilde{L}_s}{2}\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \tilde{\eta}_i^{(t)} \mathbf{h}_i^t\right)\right\|^2\right] \tag{147}$$

$$= \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2}{2}\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^N \frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}} \mathbf{h}_i^t\right)\right\|^2\right] \tag{148}$$

$$\leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right] + \underbrace{\tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t\right)\right)\right\|^2\right]}_{T_{11}}. \tag{149}$$

T_{11} here can be bounded as

$$T_{11} = \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\left(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t\right)\right)\right\|^2\right] \tag{150}$$

$$= \frac{\tilde{L}_s}{\tau^2N^2}\mathbb{E}\left[\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2\left\|\left(\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})-F_i(\mathbf{x}_i^{(t,r)})\right)\right)\right\|^2\right] \tag{151}$$

$$\leq \frac{\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2 \tag{152}$$

$$\leq \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}(\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2 + \frac{2\tilde{L}_s}{\tau N^2}\sum_{i=1}^{N}\|\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)}\|^2(\hat{b}_i^{(t)})^2\sigma^2 \tag{153}$$

$$\leq \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau N} + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}\sigma^2\epsilon_\eta}{8\tau N}. \tag{154}$$

Here equation 154 follows from lemma 11 and lemma 10. Equation (152) follows from proposition 4 and equation 151 follows from independent sampling of mini-batches by the clients. Thus we have

$$T_{10} \leq \tilde{L}_s(\tilde{\eta}^{(t)})^2\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right] + \frac{\tilde{L}_s(\tilde{\eta}^{(t)})^2w_{\max}^2\sigma^2}{8\tau N}(1+\epsilon_\eta). \tag{155}$$

Consider the term T_9

$$T_9 = -\mathbb{E}\left[\left\langle\nabla G(\bar{\mathbf{x}}^{(t,0)}),\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}\widehat{\mathbf{h}}_i^t\right)\right\rangle\right] - \mathbb{E}\left[\left\langle\nabla G(\bar{\mathbf{x}}^{(t,0)}),\left(\frac{1}{\tau N}\sum_{i=1}^{N}\tilde{\eta}_i^{(t)}(\mathbf{h}_i^t-\widehat{\mathbf{h}}_i^t)\right)\right\rangle\right] \tag{156}$$

$$= -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)})\right\|^2 - \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2}_{T_{12}} + \underbrace{\frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2}_{T_{13}}. \tag{157}$$

T_{13} her an be further bounded as

$$T_{13} = \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\nabla G(\bar{\mathbf{x}}^{(t,0)}) - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2 \tag{158}$$

$$= \frac{\tilde{\eta}^{(t)}}{2}\mathbb{E}\left\|\sum_{i=1}^{N}\left[\nabla\Psi(\bar{\mathbf{X}}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right\|^2 \tag{159}$$

$$\begin{aligned}
&\leq 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left(\left[\nabla\Psi(\bar{\mathbf{X}}^{(t,0)})\right]_i - \left[\nabla\Psi(\mathbf{X}^{(t,0)})\right]_i\right)\right\|^2\right] + 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right] \\
&\quad + 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t - \frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\right\|^2\right] + 2\tilde{\eta}^{(t)}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2\right]
\end{aligned} \tag{160}$$

$$\leq \underbrace{2\tilde{\eta}^{(t)}\tilde{L}_g^2\left\|\bar{\mathbf{X}}^{(t,0)}-\mathbf{X}^{(t,0)}\right\|_F^2}_{\text{Consensus error}} + \underbrace{2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right]}_{T_{14}}$$

$$+ \underbrace{2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t - \frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\right\|^2\right]}_{T_{15}} + \underbrace{2\tilde{\eta}^{(t)}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\widehat{\mathbf{h}}_i^t\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2\right]}_{T_{16}}. \tag{161}$$

Here equation (161) follows from Jensen's inequality and lemma 9. T_{15} and T_{16} can be bounded using lemma 14 and lemma 12 respectively

$$T_{15} = \frac{2\tilde{\eta}^{(t)}}{N^2\tau^2}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(b_i^{(t)} - \hat{b}_i^{(t)}\right)\right\|^2\right] \tag{162}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_c^2}{N\tau}\mathbb{E}\left[\sum_{r=0}^{\tau-1}\sum_{i=1}^{N}\left\|b_i^{(t)} - \hat{b}_i^{(t)}\right\|^2\right] \tag{163}$$

$$\leq 2\tilde{\eta}^{(t)}L_c^2\epsilon_b \tag{164}$$

where $K_{\max} = \max_i |\mathcal{K}_i|$. Here equation (164) follows from lemma 14. Now T_{16} can be bounded as

$$T_{16} = 2\tilde{\eta}^{(t)}\mathbb{E}\left\|\frac{1}{\tau N}\sum_{i=1}^{N}\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2 \tag{165}$$

$$\leq \frac{2\tilde{\eta}^{(t)}}{\tau N}\sum_{i=1}^{N}\sum_{r=0}^{\tau-1}\left\|\hat{b}_i^{(t)}\nabla F_i(\mathbf{x}_i^{(t,r)})\left(1-\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\right)\right\|^2 \tag{166}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_c^2}{N}\sum_{i=1}^{N}\frac{\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\|^2}{\|\tilde{\eta}^{(t)}\|^2}(\hat{b}_i^{(t)})^2 \tag{167}$$

$$\leq \frac{\tilde{\eta}^{(t)}L_c^2 w_{\max}^2}{8}\epsilon_\eta. \tag{168}$$

Equation (168) follows from lemma 12. Next we note that using the definition in equation (140) the term $\left[\Psi(\mathbf{X}^{(t,0)})\right]_i$ can be rewritten as

$$\left[\nabla\Psi(\mathbf{X}^{(t,0)})\right]_i = \frac{1}{N}\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k M_k(\mathbf{X}^{(\mathbf{t},\mathbf{0})})(1-\sigma_i(\mathbf{x}_i^{(t,0)}))\right)\nabla F_i(\mathbf{x}_i^{(t,0)})$$

$$= \frac{b_i^{(t)}}{N}\nabla F_i(\mathbf{x}_i^{(t,0)}).$$

Plugging this into T_{14} and using the definitions in equations (136),(143) we have

$$T_{14} = 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\sum_{i=1}^{N}\left[\Psi(\mathbf{X}^{(t,0)})\right]_i - \frac{1}{\tau N}\sum_{i=1}^{N}\overline{\mathbf{h}}_i^t\right\|^2\right] \tag{169}$$

$$= 2\tilde{\eta}^{(t)}\mathbb{E}\left[\left\|\frac{1}{\tau N}\sum_{i=1}^{N}b_i^{(t)}\sum_{r=0}^{\tau-1}\left(\nabla F_i(\mathbf{x}_i^{(t,0)}) - \nabla F_i(\mathbf{x}_i^{(t,r)})\right)\right\|^2\right] \tag{170}$$

$$\leq \frac{2\tilde{\eta}^{(t)}}{N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\nabla F_i(\mathbf{x}_i^{(t,0)}) - \nabla F_i(\mathbf{x}_i^{(t,r)})\right\|^2\right] \tag{171}$$

$$\leq \frac{2\tilde{\eta}^{(t)}L_s^2}{N\tau}\sum_{i=1}^{N}(b_i^{(t)})^2\sum_{r=0}^{\tau-1}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)} - \mathbf{x}_i^{(t,r)}\right\|^2\right]. \tag{172}$$

Substituting the upper bound from lemma (15) into equation (172) we can further bound T_{14} as

$$T_{14} \leq \frac{2\tilde{\eta}^{(t)}L_s^2}{N\tau}\sum_{i=1}^{N}(\hat{b}_i^{(t)})^2\left(\tau L_c^2\left((\tilde{\eta}^{(t)})^2(b_i^{(t)})^2 + \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)\right.$$
$$\left. + \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16} + 3\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]\right) \tag{173}$$

$$\leq \frac{(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 L_c^2}{128} + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2\sigma^2}{128\tau} + \frac{3\tilde{\eta}^{(t)}w_{\max}^2 L_s^2\sigma^2}{8N\tau}\sum_{i=1}^{N}E\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\right]$$

$$+ \frac{\tilde{\eta}^{(t)} w_{\max}^2 L_s^2 L_c^2}{8N} \sum_{i=1}^{N} \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\, \hat{b}_i^{(t)}\right\|^2\right] \tag{174}$$

$$\leq \frac{(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 L_c^2}{128}(1+\epsilon_\eta) + \frac{3(\tilde{\eta}^{(t)})^3 w_{\max}^4 L_s^2 \sigma^2}{128\tau}(1+\epsilon_\eta). \tag{175}$$

Here equation (174) follows from lemma 10 and equation (175) follows from lemma11. Now using equations (146), (155), (157), (161), (164), (168) and (175) we have

$$\mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] \leq -\frac{\tilde{\eta}^{(t)}}{2}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 + \tilde{\eta}^{(t)}\left(\tilde{L}_s\tilde{\eta}^{(t)} - \frac{1}{2}\right)\mathbb{E}\left[\left\|\left(\frac{1}{\tau N}\sum_{i=1}^{N}\frac{\tilde{\eta}_i^{(t)}}{\tilde{\eta}^{(t)}}\widehat{\mathbf{h}}_i^t\right)\right\|^2\right]$$
$$+ 2\tilde{\eta}^{(t)} N \tilde{L}_g^2 \Xi_t^2 + \tilde{\eta}^{(t)}\left(2L_c^2\epsilon_b + \frac{L_c^2 w_{\max}^2}{8}\epsilon_\eta\right) + (\tilde{\eta}^{(t)})^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2}{8\tau N}(1+\epsilon_\eta)\right)$$
$$+ (\tilde{\eta}^{(t)})^3\left(\frac{w_{\max}^4 L_s^2 L_c^2}{128}(1+\epsilon_\eta) + \frac{3w_{\max}^4 L_s^2\sigma^2}{128\tau}(1+\epsilon_\eta)\right). \tag{176}$$

Using $\tilde{\eta}^{(t)} \leq \frac{1}{4\tilde{L}_s}$ and $\frac{\eta}{\epsilon + \frac{N w_{\max}}{4}} \leq \tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ we can further simplify

$$\frac{\eta}{2\left(\epsilon + \frac{N w_{\max}}{4}\right)}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 \leq \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t,0)})\right] - \mathbb{E}\left[G(\bar{\mathbf{x}}^{(t+1,0)})\right] + \frac{2\eta N \tilde{L}_g^2 \Xi_t^2}{\epsilon} + \eta^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2 (1+\epsilon_\eta)}{8\tau N \epsilon^2}\right)$$
$$+ \eta\left(\frac{2L_c^2\epsilon_b}{\epsilon} + \frac{L_c^2 w_{\max}^2}{8\epsilon}\epsilon_\eta\right) + \eta^3\left(\frac{w_{\max}^4 L_s^2 L_c^2}{128\epsilon^3}(1+\epsilon_\eta) + \frac{3w_{\max}^4 L_s^2\sigma^2}{128\tau\epsilon^3}(1+\epsilon_\eta)\right). \tag{177}$$

Averaging over T and using lemma 16 alongwith the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{\epsilon}$ we have

$$\frac{\eta}{2\left(\epsilon + \frac{N w_{\max}}{4}\right)}\frac{1}{T}\sum_{t=1}^{T}\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2 \leq \frac{G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}}{T} + \eta^2\left(\frac{\tilde{L}_s w_{\max}^2 \sigma^2 (1+\epsilon_\eta)}{8\tau N\epsilon^2}\right) + \eta\left(\frac{4N\tilde{L}_g\Xi_0^2}{\epsilon p T} + \frac{2L_c^2\epsilon_b}{\epsilon} + \right.$$
$$\left. \frac{L_c^2 w_{\max}^2}{8\epsilon}\epsilon_\eta\right) + \eta^3\left(\frac{w_{\max}^4 L_s^2 L_c^2(1+\epsilon_\eta)}{128\epsilon^3} + \frac{3w_{\max}^4 L_s^2\sigma^2(1+\epsilon_\eta)}{128\tau\epsilon^3} + \frac{3N\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)}{2p^2T\epsilon^3} + \frac{3N\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)}{p^2\tau T\epsilon^3}\right). \tag{178}$$

Choosing $\epsilon = \frac{\gamma N B_1}{L_s}$ where $B_1 = \frac{\lambda_{\max} C_{\max} K_{\max} L_c^2}{16} + \frac{\lambda_{\max} K_{\max} L_c^2}{6\sqrt{3}}$(see lemma 8) to make the step size inversely proportional to the global smoothness parameter

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G(\overline{\mathbf{x}}^{(t,0)})\right\|^2\right] \leq \frac{A_0}{\eta T} + A_1 + A_2\eta + A_3\eta^2 \tag{179}$$

where

$$A_0 = 2N\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[G(\overline{\mathbf{x}}^{(0,0)}) - G_{\inf}\right], \tag{180}$$

$$A_1 = 2\left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right)\left(\frac{4N\tilde{L}_g\Xi_0^2}{\gamma B_1 p T} + \frac{2L_c^2\epsilon_b}{\gamma B_1} + \frac{L_c^2 w_{\max}^2}{8\gamma B_1}\epsilon_\eta\right), \tag{181}$$

$$A_2 = \frac{\tilde{L}_s\, w_{\max}^2\sigma^2(1+\epsilon_\eta)L_s}{4\tau N^2\gamma^2 B_1^2}\left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right), \tag{182}$$

$$A_3 = \left(\gamma B_1 + \frac{L_s w_{\max}}{4}\right)\left[\frac{w_{\max}^4 L_s^4 L_c^2(1+\epsilon_\eta)}{64N^2\gamma^3 B_1^3} + \frac{3w_{\max}^4 L_s^4\sigma^2(1+\epsilon_\eta)}{64\tau N^2\gamma^3 B_1^3}\right.$$
$$\left. + \frac{3\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)L_s^2}{p^2 T N\gamma^3 B_1^3} + \frac{6\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)L_s^2}{p^2\tau T N\gamma^3 B_1^3}\right]. \tag{183}$$

For small values of gamma we can use the fact $\tilde{\eta}^{(t)} \leq \frac{\eta}{N(1-\gamma)}$ and using the same steps as above starting from equation (45) we obtain the bound

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq \frac{A_0}{\eta T} + A_1 + A_2\eta + A_3\eta^2 \tag{184}$$

for

$$A_0 = 2N\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[G(\overline{\mathbf{x}}^{(0,0)}) - G_{\inf}\right], \tag{185}$$

$$A_1 = 2\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left(\frac{4N\tilde{L}_g\Xi_0^2}{(1-\gamma)pT} + \frac{2L_c^2\epsilon_b}{(1-\gamma)} + \frac{L_c^2 w_{\max}^2}{8(1-\gamma)}\epsilon_\eta\right), \tag{186}$$

$$A_2 = \frac{\tilde{L}_s\, w_{\max}^2\sigma^2(1+\epsilon_\eta)}{4\tau N^2(1-\gamma)^2}\left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right) \tag{187}$$

$$A_3 = \left(\frac{\gamma B_1}{L_s} + \frac{w_{\max}}{4}\right)\left[\frac{w_{\max}^4 L_s^2 L_c^2(1+\epsilon_\eta)}{64N^2(1-\gamma)^3} + \frac{3w_{\max}^4 L_s^2\sigma^2(1+\epsilon_\eta)}{64\tau N^2(1-\gamma)^3}\right.$$
$$\left. + \frac{3\tilde{L}_g^2 w_{\max}^2 L_c^2(1+\epsilon_\eta)}{p^2TN(1-\gamma)^3} + \frac{6\tilde{L}_g^2 w_{\max}^2\sigma^2(1+\epsilon_\eta)}{p^2\tau TN(1-\gamma)^3}\right]. \tag{188}$$

The final bound is obtained by taking the minimum of the above two expressions and using lemma 1. Specifically choosing $A_4 = \max\left\{\frac{4\tilde{L}_s}{\epsilon}, \frac{3}{\sqrt{2}\epsilon}\right\}$ we have for $\epsilon = \frac{\gamma N B_1}{L_s}$ chosen as per lemma (8).

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla G\left(\overline{\mathbf{x}}^{(t,0)}\right)\right\|^2\right] \leq A_1 + 2\left(\frac{A_2A_0}{T}\right)^{\frac{1}{2}} + 2A_3^{\frac{1}{3}}\left(\frac{A_0}{T}\right)^{\frac{2}{3}} + \frac{A_4A_0}{T} \tag{189}$$

$$= \mathcal{O}\left(\epsilon_\eta\right) + \mathcal{O}\left(\epsilon_b\right) + \mathcal{O}\left(\sqrt{\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)(1+\epsilon_\eta)}{N\tau T}}\right)$$
$$+ \mathcal{O}\left(\frac{N\Xi_0^2}{pT}\right) + \mathcal{O}\left(\frac{(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf})}{T}\right)$$
$$+ \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}}{T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) + \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}N^{1/3}}{p^{2/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right)$$
$$+ \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}}{\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) + \mathcal{O}\left(\frac{\left(G(\bar{\mathbf{x}}^{(0,0)}) - G_{\inf}\right)^{2/3}N^{1/3}}{p^{2/3}\tau^{1/3}T^{2/3}}(1+\epsilon_\eta)^{1/3}\right) \tag{190}$$

D.1 Auxiliary Lemmas

LEMMA 8 (SMOOTHNESS OF THE GLOBAL OBJECTIVE). *Under Assumption 2 the smoothness parameter of the global objective $G(\mathbf{x})$ for any $\mathbf{x}$ is given by $\gamma B_1 + \frac{L_s}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right)$ where $B_1 = \frac{\lambda_{\max}C_{\max}K_{\max}L_c^2}{16} + \frac{\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}$.*

PROOF. From equation (133) have

$$\nabla G(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^N\left((1-\gamma) + \gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right)\nabla F_i(\mathbf{x}). \tag{191}$$

Let $\nabla G_1(\mathbf{x}) := \frac{(1-\gamma)}{N}\sum_{i=1}^N F_i(\mathbf{x})$ and $\nabla G_2(\mathbf{x}) := \frac{\gamma}{N}\sum_{i=1}^N\left\{\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x})\nabla F_i(\mathbf{x})\right\}$. We have

$$\left\|\nabla^2 G_2(\mathbf{x})\right\|_{op} = \gamma\left\|\frac{1}{N}\sum_{i=1}^N\sum_{k\in\mathcal{K}_i}\lambda_k\left\{\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\nabla^2F_i(\mathbf{x})\right.\right.$$
$$+ \sum_{\substack{l\in C_k\\ i\neq l}}\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_l(\mathbf{x}))(1-\sigma_i(\mathbf{x}))\nabla F_l(\mathbf{x})\nabla F_i(\mathbf{x})^T$$

$$\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))(1-2\sigma_i(\mathbf{x}))\nabla F_i(\mathbf{x})\nabla F_i(\mathbf{x})^T\Bigg\|_{op}. \tag{192}$$

Using triangle inequality for operator norms and $\sigma(x)(1-\sigma(x)) \le 1/4$ and $\sigma(x)(1-\sigma(x))(1-2\sigma(x)) \le 1/6\sqrt{3}$ we have

$$\left\|\nabla^2 G_2(\mathbf{x})\right\|_{op} \le \frac{\gamma}{N}\sum_{i=1}^{N}\sum_{k\in\mathcal{K}_i}\lambda_k\left\{L_s\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))+\sum_{\substack{l\in C_k\\ i\neq l}}\frac{L_c^2}{16}+\frac{L_c^2}{6\sqrt{3}}\right\} \tag{193}$$

$$\le \frac{\gamma\lambda_{\max}C_{\max}K_{\max}L_c^2}{16}+\frac{\gamma\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}+\frac{\gamma L_s}{N}\sum_{i=1}^{N}\left\{\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right\}. \tag{194}$$

Also $\|\nabla^2 G_1(\mathbf{x})\|_{op} \le (1-\gamma)L_s$. Since $\|\nabla^2 G(\mathbf{x})\|_{op} \le \|\nabla^2 G_1(\mathbf{x})\|_{op} + \|\nabla^2 G_2(\mathbf{x})\|_{op}$

$$\|\nabla^2 G(\mathbf{x})\|_{op} \le \gamma B_1 + \frac{L_s}{N}\sum_{i=1}^{N}\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x})\right)(1-\sigma_i(\mathbf{x}))\right) \tag{195}$$

where $B_1 = \frac{\lambda_{\max}C_{\max}K_{\max}L_c^2}{16} + \frac{\lambda_{\max}K_{\max}L_c^2}{6\sqrt{3}}$. □

LEMMA 9 (SMOOTHNESS OF Ψ). *For all $i \in [N]$ we have $\|[\nabla\Psi(\mathbf{X})]_i - [\Psi(\mathbf{Y})]_i\|_2^2 \le \frac{\tilde{L}_g}{N}\|\mathbf{X}-\mathbf{Y}\|_F^2$ where $\tilde{L}_g = \frac{L_s w_{\max}}{4} + \frac{L_c^2 w_{\max}}{4}$.*

PROOF. From equation (136) we have

$$[\nabla\Psi(\mathbf{X})]_i = \nabla_{\mathbf{x}_i}\Psi(\mathbf{X}) = \frac{1}{N}\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i). \tag{196}$$

We also have

$$\nabla^2_{\mathbf{x}_i}\Psi(\mathbf{X}) = \frac{1}{N}\left\{\left((1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))\right)\nabla^2 F_i(\mathbf{x}_i)\right. \tag{197}$$

$$\left.+\left(\sum_{k\in\mathcal{K}_i}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))(1-2\sigma_i(\mathbf{x}_i))\right)\nabla F_i(\mathbf{x}_i)\nabla F_i(\mathbf{x}_i)^\top\right\} \tag{198}$$

and for $i \neq l$ we have

$$\nabla^2_{\mathbf{x}_i,\mathbf{x}_l}\Psi(\mathbf{X}) = \frac{1}{N}\sum_{k\in\mathcal{K}_i\cap\mathcal{K}_l}\lambda_k\left(\prod_{j\in C_k}\sigma_j(\mathbf{x}_j)\right)(1-\sigma_i(\mathbf{x}_i))(1-\sigma_l(\mathbf{x}_l))\nabla F_i(\mathbf{x}_i)\nabla F_l(\mathbf{x}_l)^\top. \tag{199}$$

Using triangle inequality for operator norms and $\sigma(x)(1-\sigma(x)) \le 1/4$ and $\sigma(x)(1-\sigma(x))(1-2\sigma(x)) \le 1/6\sqrt{3}$ we have

$$\|\nabla^2_{\mathbf{x}_i}\Psi(\mathbf{X})\|_{op} \le \frac{L_s(4(1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k)}{4N}+\frac{L_c^2(6\sqrt{3}(1-\gamma)+\gamma\sum_{k\in\mathcal{K}_i}\lambda_k)}{6\sqrt{3}N} \le \frac{L_s w_{\max}}{4N}+\frac{L_c^2 w_{\max}}{4N} \tag{200}$$

$$\|\nabla^2_{\mathbf{x}_i,\mathbf{x}_l}\Psi(\mathbf{X})\|_{op} \le \frac{L_c^2(\sum_{k\in\mathcal{K}_i\cap\mathcal{K}_j}\lambda_i)}{6\sqrt{3}N}. \tag{201}$$

Now by the mean value theorem for any $\mathbf{X}$ and $\mathbf{Y}$, we have $[\nabla\Psi(\mathbf{X})]_i - [\nabla\Psi(\mathbf{Y})]_i = \sum_{j=1}^N \nabla^2_{x_i,x_j}\Psi(\mathbf{Z})(x_j-y_j)$ for some intermediate point $\mathbf{Z} = t\mathbf{X}+(1-t)\mathbf{Y}$ where $t\in[0,1]$. A uniform upper bound on these operator norms for all i can be given by $\tilde{L}_g/N$ where $\tilde{L}_g = \frac{L_s w_{\max}}{4}+\frac{L_c^2 w_{\max}}{4}$ and thus $\|\nabla^2_{x_i,x_j}\Psi(\mathbf{Z})\|_{op} \le \tilde{L}_g/N$, we obtain $\|[\nabla\Psi(\mathbf{X})]_i - [\nabla\Psi(\mathbf{Y})]_i\| \le \tilde{L}_g/N\sum_{j=1}^N\|x_j-y_j\|$. Using Jensen's inequality we obtain

$$\|[\nabla\Psi(\mathbf{X})]_i - [\Psi(\mathbf{Y})]_i\|_2^2 \le \frac{\tilde{L}_g^2}{N}\sum_{j=1}^{N}\|x_j-y_j\|^2. \tag{202}$$

□

LEMMA 10 (PRE-FACTOR UPPER BOUND). *For $w_{\max} = 4(1-\gamma)+\gamma\max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)$, we have $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ and $b_i^{(t)} \le \frac{w_{\max}}{4}$ $\forall i\in[N]$ and t.*

PROOF. We have

$$\hat{b}_i^{(t)} = (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k M_{ki}(\mathbf{X}^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})).$$

By definition, it follows that

$$\left(M_{ki}^{(0)}(\mathbf{X}^{(t,0)})\right)^{1/N} = \sigma_i(\mathbf{x}_i^{(t,0)})$$

and

$$M_{ki}(\mathbf{X}^{(t,0)}) = \min\left\{\sigma_i(\mathbf{x}_i^{(t,0)}), M_{ki}^{(\tau_b)}(\mathbf{X}^{(t,0)})\right\}.$$

Therefore, we obtain the bound

$$\hat{b}_i^{(t)} \le (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k \sigma_i(\mathbf{x}_i^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})) \le (1-\gamma) + \frac{\gamma \sum_{k\in\mathcal{K}_i}\lambda_k}{4} \le \frac{4(1-\gamma) + \gamma \max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)}{4}.$$

We also have $i \in \mathcal{K}_i$ by definition thus $M_k(\mathbf{X}^{(t,0)}) \le \sigma_i(\mathbf{x}_\mathbf{i}^{(\mathbf{t},\mathbf{0})})$ and

$$b_i^{(t)} \le (1-\gamma) + \gamma \sum_{k\in\mathcal{K}_i} \lambda_k \sigma_i(\mathbf{x}_i^{(t,0)})(1-\sigma_i(\mathbf{x}_i^{(t,0)})) \le (1-\gamma) + \frac{\gamma \sum_{k\in\mathcal{K}_i}\lambda_k}{4} \le \frac{4(1-\gamma) + \gamma \max_{i\in[N]}\left(\sum_{k\in\mathcal{K}_i}\lambda_k\right)}{4}.$$

□

LEMMA 11 (LEARNING RATE ESTIMATION ERROR - GROUPWISE PENALTY). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2 (\hat{b}_i^{(t)})^2 \le \min\left\{\frac{(\eta)^2(1-p)^{\tau_\eta}N^2}{4}, \frac{w_{\max}^2(\tilde{\eta}^{(t)})^2(1-p)^{\tau_\eta}N^2}{64}\right\} \tag{203}$$

$$= \min\left\{(\eta)^2\epsilon_\eta, \frac{(\tilde{\eta}^{(t)})^2 w_{\max}^2 \epsilon_\eta}{16}\right\}. \tag{204}$$

where $\epsilon_\eta = \frac{(1-p)^{\tau_\eta}N^2}{4}$. Note that ϵ_η can be made arbitrarily small as required by increasing τ_η.

PROOF. We have $\tilde{\eta}_i^{(t)} = \frac{\eta}{s_i^{(t)}+\epsilon}$ and $\tilde{\eta}^{(t)} = \frac{\eta}{\bar{s}^{(t)}+\epsilon}$ where $s_i^{(t)} = \max\{s_i^{(t,\tau_\eta)}, s_i^{(t,0)}/N\}$, $s_i^{(t,0)} = N\hat{b}_i^{(t)}$, and $s_i^{(t,r+1)} = \sum_{j\in\mathcal{N}_i} w_{ij}\, s_j^{(t,r)}$, or equivalently $\mathbf{s}^{(t,r+1)} = \mathbf{W}\mathbf{s}^{(t,r)}$, with $\mathbf{s}^{(t,r)} = [s_1^{(t,r)} \cdots s_N^{(t,r)}]$. The proof follows same as lemma 3 after replacing $b_i^{(t)}$ by $\hat{b}_i^{(t)}$ and using $\hat{b}_i^{(t)} \le \frac{w_{\max}}{4}$ from lemma 10. □

LEMMA 12 (FRACTIONAL LEARNING RATE ESTIMATION ERROR - GROUPWISE PENALTY). *For τ_η rounds of gossip averaging used for estimating $s_i^{(t)}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\tilde{\eta}_i^{(t)} - \tilde{\eta}^{(t)}\right\|^2}{\|\tilde{\eta}^{(t)}\|^2}(\hat{b}_i^{(t)})^2 \le \frac{(1-p)^{\tau_\eta}N^2(\max_i(\hat{b}_i^{(t)}))^2}{4} = \max_i(\hat{b}_i^{(t)})^2\epsilon_\eta \le \frac{w_{\max}^2\epsilon_\eta}{16}. \tag{205}$$

PROOF. Proceeding similarly as lemma 3 replacing $b_i^{(t)}$ by $\hat{b}_i^{(t)}$ and using lemma 10 for bounding $\hat{b}_i^{(t)}$ completes the proof. □

LEMMA 13 (PRODUCT ESTIMATION ERROR). *If $M_{ki}(\mathbf{X})$ is the estimate of $M_k(\mathbf{X})$ at node i computed as per equation* (137)*, then for any vector $\mathbf{X}$ we have*

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\,\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le (1-p)^{\tau_b}(N-1)\sum_{k=1}^{K}\lambda_k^2 = \epsilon_M. \tag{206}$$

Note that this term can be made arbitrarily small by increasing τ_b.

PROOF. We have $M_k(\mathbf{X}) := \prod_{j\in C_k}\sigma_j(\mathbf{x}_j)$ and $M_{ki}(\mathbf{X}) = \min\left\{M_{ki}^{(\tau_b)}(\mathbf{X}), (M_{ki}^{(0)}(\mathbf{X}))^{1/N}\right\}$. Note that $M_k(\mathbf{X}) \le (M_{ki}^{(0)}(\mathbf{X}))^{1/N}$ $\forall i$ and therefore $\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le \left\|M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}(\mathbf{X})\right\|^2$. Now we have

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\,\|M_k(\mathbf{X}) - M_{ki}(\mathbf{X})\|^2 \le \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\lambda_k^2\left\|M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}\right\|^2 \tag{207}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2\left\|M_k(\mathbf{X}) - M_k(\mathbf{X})^{\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}}\right\|^2 \tag{208}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| M_k(\mathbf{X}) - M_k(\mathbf{X})^{1+\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}} \right\|^2. \tag{209}$$

For a positive ϵ the inequality, $a - a^{1+\epsilon} \leq \frac{\epsilon}{(1+\epsilon)^{1+\frac{1}{\epsilon}}} \leq \frac{\epsilon}{1+\epsilon} \leq \epsilon$ holds for $a \in [0,1]$. Since both $M_k(\mathbf{X})$ and $M_{ki}^{(\tau_b)}(\mathbf{X}) \in [0,1]$, if $\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} > 0$ in equation(209) for $\epsilon = \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}$

$$\frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| M_k(\mathbf{X}) - M_k(\mathbf{X})^{1+\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))}} \right\|^2 \leq \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2. \tag{210}$$

For the case where $\frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} < 0$ we can do the same procedure using $M_{ki}^{(\tau_b)}(\mathbf{X})$ as the base, here the ϵ would be $\frac{\ln(M_k(\mathbf{X}))-\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))}{\ln(M_{ki}(\mathbf{X}))}$ and we upper bound using $\frac{\epsilon}{1+\epsilon}$ to obtain $\left\| M_k(\mathbf{X}) - M_{ki}^{(\tau_b)}(\mathbf{X}) \right\|^2 \leq \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X}))-\ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2$. Since the numerator in this expression corresponds exactly to the squared norm of a scaled version of the object being computed over the network via gossip. from assumption 4, after τ_b rounds of gossip with the mixing matrix $\mathbf{W}$ we obtain

$$\frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \frac{\ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2 = \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left(\frac{\sum_{i=1}^{N}\left\| \ln(M_{ki}^{(\tau_b)}(\mathbf{X})) - \ln(M_k(\mathbf{X})) \right\|^2}{\|\ln(M_k(\mathbf{X}))\|^2} \right) \tag{211}$$

$$\leq \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left(\frac{(1-p)^{\tau_b}\sum_{i=1}^{N}\left\| \ln(M_{ki}^{(0)}(\mathbf{X})) - \ln(M_k(\mathbf{X})) \right\|^2}{\|\ln(M_k(\mathbf{X}))\|^2} \right) \tag{212}$$

$$= \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 \left((1-p)^{\tau_b}\sum_{i=1}^{N}\left\| \frac{\ln(M_{ki}^{(0)}(\mathbf{X})) - \ln(M_k(\mathbf{X}))}{\ln(M_k(\mathbf{X}))} \right\|^2 \right) \tag{213}$$

$$\leq \frac{1}{N}\sum_{k=1}^{K}\lambda_k^2 (1-p)^{\tau_b}(N^2 - N) \tag{214}$$

$$= (1-p)^{\tau_b}(N-1)\sum_{k=1}^{K}\lambda_k^2 = \epsilon_M. \tag{215}$$

Additionally, Equation (213) exhibits a structure similar to the coefficient of variation. Equation (215) follows from the fact that for non-negative weights w_i and the mean weight $w = \frac{1}{N}\sum_{i=1}^{N} w_i$, the following inequality holds:

$$\sum_{i=1}^{N}\frac{\|w_i - w\|^2}{\|w\|^2} = \frac{\sum_{i=1}^{N}\|w_i\|^2}{\|\frac{1}{N}\sum_{i=1}^{N} w_i\|^2} - N \leq N^2 - N.$$

□

LEMMA 14 (PRE-FACTOR ESTIMATION ERROR). *The pre-factor estimation error after τ_b rounds of gossip averaging used for estimating the product terms $M_k(\mathbf{X}^{(t,0)})$ can be upper bounded as*

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| b_i^{(t)} - \hat{b}_i^{(t)} \right\|^2 \right] \leq K_{\max}\epsilon_M = \epsilon_b. \tag{216}$$

PROOF.

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| b_i^{(t)} - \hat{b}_i^{(t)} \right\|^2 \right] = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\left\| \sum_{k\in\mathcal{K}_i}\lambda_k \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 (1-\sigma_i(\mathbf{x}_i^{(t,0)}))^2 \right] \tag{217}$$

$$\leq \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\sum_{k\in\mathcal{K}_i}|\mathcal{K}_i|\lambda_k^2 \left\| \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 \right] \tag{218}$$

$$\leq \frac{K_{\max}}{N}\mathbb{E}\left[\sum_{k=1}^{K}\sum_{i=1}^{N}\lambda_k^2 \left\| \left(M_k(\mathbf{X}^{(t,0)}) - M_{ki}(\mathbf{X}^{(t,0)}) \right) \right\|^2 \right] \tag{219}$$

$$\leq K_{\max}\epsilon_M = \epsilon_b \tag{220}$$

□

LEMMA 15 (BOUNDING DRIFT - GROUPWISE PENALTY). *For client i, the accumulated local drift error over τ local steps in Algorithm 3, can be bounded as:*

$$\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq \tau L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right) \frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2\sigma^2}{16}+3\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{221}$$

PROOF. Using the expression for the accumulated local update after r local steps:

$$\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] = \mathbb{E}\left[\left\|\tilde{\eta}_i^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right\|^2\right] \tag{222}$$

$$\leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+3\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + \underbrace{3\mathbb{E}\left[\left\|(\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right)\right\|^2\right]}_{T_{17}}. \tag{223}$$

Consider T_{17}

$$T_{17} = 3\mathbb{E}\left[\left\|(\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\left(\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right)\right\|^2\right] \tag{224}$$

$$\leq 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2\,(\hat{b}_i^{(t)})^2\sum_{l=0}^{r-1}\left\|\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)-\nabla F_i\left(\mathbf{x}_i^{(t,l)},\xi_i^{(t,l)}\right)\right\|^2\right] \tag{225}$$

$$\leq 3\mathbb{E}\left[(\tilde{\eta}_i^{(t)})^2\,(\hat{b}_i^{(t)})^2 r\sigma^2\right] \tag{226}$$

$$\leq 6\mathbb{E}\left[(\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2 r\sigma^2\right]+6\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2 r\sigma^2\right] \tag{227}$$

$$\leq \frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{228}$$

Here equation (225) follows from proposition 4 and equation 228 follows from lemma 10. Plugging this back into equation 223 we have

$$\tau^2\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq 3\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right] + 3\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\sum_{l=0}^{r-1}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+\frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right] \tag{229}$$

$$\leq 3r\sum_{l=1}^{r-1}\left(\mathbb{E}\left[\left\|\tilde{\eta}^{(t)}\,\hat{b}_i^{(t)}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\nabla F_i\left(\mathbf{x}_i^{(t,l)}\right)\right\|^2\right]\right) + \frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right] \tag{230}$$

$$\leq 3r^2L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)+\frac{3(\tilde{\eta}^{(t)})^2 r w_{\max}^2\sigma^2}{8}+6r\mathbb{E}\left[(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]. \tag{231}$$

Summing over r and dividing by τ^2 we have

$$\sum_{r=1}^{\tau}\mathbb{E}\left[\left\|\mathbf{x}_i^{(t,0)}-\mathbf{x}_i^{(t,r)}\right\|^2\right] \leq \tau L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2+\mathbb{E}\left[\left\|(\tilde{\eta}^{(t)}-\tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right)$$

$$\frac{3(\tilde{\eta}^{(t)})^2 w_{\max}^2 \sigma^2}{16} + 3\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2 (\hat{b}_i^{(t)})^2 \sigma^2\right]. \tag{232}$$

□

LEMMA 16 (CONSENSUS CONTROL- GROUPWISE PENALTY). *The term* $\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2$ *can be bounded as*

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \leq \frac{2\Xi_0^2}{pT} + \left(\frac{3w_{\max}^2 L_c^2}{4p^2 T} + \frac{3w_{\max}^2\sigma^2}{2p^2\tau T}\right)(1+\epsilon_\eta)\sum_{t=0}^{T-1}(\tilde{\eta}^{(t)})^2. \tag{233}$$

(234)

PROOF. Proceeding similar to the proof of lemma 6 replacing $q_i(\mathbf{x}_i^{(t,0)})a_i$ by $\hat{b}_i^{(t)}$ we have

$$N\Xi_{t+1}^2 \leq \left(1-\frac{p}{2}\right)N\Xi_t^2 + \underbrace{\frac{2}{p\tau^2}\sum_{i=1}^{N}\mathbb{E}\left\|\tilde{\eta}_i^{(t)}\hat{b}_i^{(t)}\sum_{r=0}^{\tau-1}\nabla F_i(\mathbf{x}_i^{(t,r)},\xi_i^{(t,r)})\right\|^2}_{T_{18}}. \tag{235}$$

Following the same steps as equations (222)-(231), replacing r with τ we bound T_{18} as

$$\begin{aligned} T_{18} \leq \frac{2}{p\tau^2}\sum_{i=1}^{N}\Bigg(&3\tau^2 L_c^2\left((\tilde{\eta}^{(t)})^2(\hat{b}_i^{(t)})^2 + \mathbb{E}\left[\left\|(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})\,\hat{b}_i^{(t)}\right\|^2\right]\right) \\ &+ \frac{3(\tilde{\eta}^{(t)})^2\tau w_{\max}^2\sigma^2}{8} + 6\tau\mathbb{E}\left[(\tilde{\eta}^{(t)} - \tilde{\eta}_i^{(t)})^2(\hat{b}_i^{(t)})^2\sigma^2\right]\Bigg). \end{aligned} \tag{236}$$

Now using lemma 10 and 11 we can bound (236) as

$$T_{18} \leq \frac{3(\tilde{\eta}^{(t)})^2 N w_{\max}^2 L_c^2}{8p}(1+\epsilon_\eta) + \frac{3(\tilde{\eta}^{(t)})^2 N w_{\max}^2\sigma^2}{4p\tau}(1+\epsilon_\eta) = NA_t. \tag{237}$$

Thus we have

$$\Xi_{t+1}^2 \leq \left(1-\frac{p}{2}\right)\Xi_t^2 + A_t. \tag{238}$$

Rearranging and summing over T we get

$$\Xi_T^2 - \Xi_0^2 = \sum_{t=0}^{T-1}\left(\Xi_{t+1}^2 - \Xi_t^2\right) \leq -\frac{p}{2}\sum_{t=0}^{T-1}\Xi_t^2 + \sum_{t=0}^{T-1}A_t. \tag{239}$$

Since $\Xi_T^2 \geq 0$

$$\frac{1}{T}\sum_{t=0}^{T-1}\Xi_t^2 \leq \frac{2}{pT}\left(\Xi_0^2 + \sum_{t=0}^{T-1}A_t\right). \tag{240}$$

Substituting A_t completes the proof. □

E Additional experimental details

Choosing a_i and γ: Note that a scaled version of our objective function can be rewritten as $F(\mathbf{x}) + \beta\tilde{F}(\mathbf{x})$ where $\tilde{F}(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N} a_i\sigma_i(\mathbf{x})$ and $\beta = \frac{\gamma}{1-\gamma}$. For all our experiments $\beta = 20$. A large value for β can be used here because the sigmoid function decays exponentially once a client's loss falls below the threshold and the objective is dominated by F. As the heterogeneity in data increases we typically need larger values of β to retain more clients. In real-world scenarios, we often have some prior knowledge about the importance of retaining node i in the system. This importance may reflect the amount or quality of data the node contributes, or the cost and complexity of restoring network connectivity if the node is lost. Such information can serve as a useful proxy when selecting the values of a_i to incorporate the cost of losing client i into the objective function, as shown by the results.

Hyperparameters and local steps: For all our experiments we set a_i to be the degree of node i. For all settings, we perform a grid search over the relevant hyperparameters to find the best performing ones. Specifically, we grid search over the learning rate for the decentralized baseline $\eta \in \{0.05, 0.01, 0.005, 0.001\}$, batch size utilized was 32$\in \{32, 64\}$ and number of local steps $\tau = \{5, 10\}$. We set the number of gossip averaging steps used to estimate the scaling factors (τ_η) and pre-factors (τ_b) to 5 in all settings. The reported results are averaged over 10 random seeds for all experiments.

Choosing client Thresholds: A practical approach to selecting local loss thresholds for each client involves running a few warm-up steps of local SGD at client k to obtain the model $\widehat{\mathbf{x}}_k$, then using the training loss $F(\widehat{\mathbf{x}}_k)$ of the resulting solo trained model obtained as the threshold. This method provides a data-driven baseline that reflects each client's local data distribution and optimization dynamics. While thresholds can be tailored to client-specific preferences or constraints in real-world settings, for our experiments, we adopt a uniform strategy: each client

performs 100 warmup steps to establish its threshold. This ensures consistency across clients while still capturing their unique local training behaviors.